Quantum Entanglement

Leonard Susskind

Supplementary track, 2006 Fall Part 3
Transcript-derived notes curated by LazyingArt LLC

Original lectures by Leonard Susskind. Transcript-derived course notes curated by LazyingArt LLC.

Contents

1 Special Relativity: Frames, Simultaneity, and the Lorentz Transformation **1**

1.1 A Short Framing Note . . . 1

1.2 Reference Frames as Lattices of Meter Sticks and Clocks . . . 1

1.3 Galilean Relativity from the Spacetime Picture . . . 2

1.4 What Galileo Gets Right . . . 3

1.5 Maxwell's Challenge and Einstein's Return to First Principles . . . 5

1.5.1 Question & Answer . . . 6

1.6 The Geometry of Relativity of Simultaneity . . . 6

1.7 From Tilted Axes to the Lorentz Transformation . . . 9

1.8 Summary . . . 13

2 Proper Time, Time Dilation, and Length Contraction **14**

2.1 Review: Galilean Relativity and Its Spacetime Grid . . . 14

2.2 From Galileo to Lorentz . . . 15

2.2.1 Question & Answer . . . 16

2.3 Light-Cone Coordinates and the Geometry of Lorentz Transformations . . . 16

2.4 Proper Time as the Invariant Interval . . . 17

2.5 Time Dilation and Relativity of Simultaneity . . . 19

2.5.1 Question & Answer . . . 20

2.6 Twin Paradox, Ideal Clocks, and Accelerated Motion . . . 21

2.6.1 Question & Answer . . . 22

2.7 Signals, Closed Spatial Loops, and Length Contraction . . . 23

2.7.1 Question & Answer . . . 23

2.8 Summary . . . 26

3 From Heavy-Ion Fireballs to Four-Vectors **27**

3.1 Heavy-Ion Collisions, Horizons, and the Fifth-Dimensional Picture . . . 27

3.2 Collider Energetics and Why Beam Geometry Matters 28
3.3 Relativity of Simultaneity Through a Spacetime Diagram 29
3.3.1 Question & Answer 29
3.4 Four Coordinates, Covariant Notation, and the Invariant Interval 30
3.5 Rotations, Matrices, and the Structure of Lorentz Transformations 31
3.6 Proper Velocity as a Four-Vector and as a Tangent Vector 34
3.6.1 Question & Answer 34
3.7 Momentum, Energy, and the Threshold of Mechanics 38
3.8 Summary 39

4 Maxwell Equations, Covariance, and Electromagnetic Waves 40
4.1 From the Lorentz Force to the Missing Field Equations 40
4.2 Why Charges Must Affect Fields 41
4.2.1 Question & Answer 41
4.3 Wave Review and Vector-Calculus Review 42
4.4 Vacuum Maxwell Equations and the Relativity Puzzle 43
4.4.1 Question & Answer 44
4.5 The Electromagnetic Field Tensor 44
4.6 From Maxwell to Electromagnetic Waves 46
4.6.1 Question & Answer 46
4.7 Restoring Sources: Charge, Current, Continuity, and Covariance 48
4.7.1 Question & Answer 48
4.8 Summary 50

Chapter 1

Special Relativity: Frames, Simultaneity, and the Lorentz Transformation

These notes follow Leonard Susskind's lecture, with curation for this edition by LazyingArt LLC. The lecture opens by setting the agenda for the quarter: first special relativity done carefully as kinematics, then relativistic dynamics, and only later a more conceptual pass through general relativity. We therefore begin where the lecture itself begins in earnest, with the meaning of a frame of reference, with clocks and meter sticks, and with the insistence that the right way to start a relativity problem is to draw spacetime.

1.1 A Short Framing Note

Relativity, in the sense used here, is the statement that the laws of physics should not depend on the uniform motion of the observer. That idea is older than Einstein. Susskind likes to make the point with a train moving so smoothly that nothing inside it rattles or shakes. In such a train, the laws of juggling are the same as they are in a train standing at the station. If we throw a ball upward, we do not have to compensate for the train's motion in order to catch it again. The physics inside the smoothly moving train is the same physics.

What Einstein changes is not the relativity principle itself, but the structure of space and time needed to preserve it once light is brought into the story. Newton imagined a universal time, almost a God's time, common to all clocks everywhere. Einstein keeps the relativity principle and abandons the universality of simultaneity. The lecture proceeds exactly in that order: first Galileo, then the crisis created by light, then the reconstruction of space and time.

1.2 Reference Frames as Lattices of Meter Sticks and Clocks

Definition 1.1. A reference frame is an imaginary lattice of meter sticks filling space, with a clock at each lattice point. An event is a point in spacetime labeled by a spatial location in that lattice and by the reading of the appropriate clock.

Figure 1.1: Rest-frame time axis and the emerging spacetime lattice.

The lecture immediately turns this operational definition into a picture. Time is drawn vertically, space horizontally, and for the moment we suppress the y and z directions and work only with one spatial coordinate x. Every point on the board then represents a place together with a time.

The horizontal axis is the set of events with $t = 0$. The vertical line through the origin is the worldline of the point $x = 0$. The next vertical line is the worldline of the point $x = 1$, then $x = 2$, and so on. As the clocks tick upward in time, the meter-stick lattice produces a family of constant-position worldlines. When we add horizontal lines of constant time, the whole picture becomes a spacetime lattice.

The board at this point is still sparse, and that is important. Susskind is not yet racing toward formulas. He is teaching us how to think about a frame: as a grid of locations and synchronized clocks, made visible in spacetime before any transformation law is written down.

1.3 Galilean Relativity from the Spacetime Picture

The lecture now brings back the train picture and makes it geometrical. Let the stationary observer use coordinates (x, t), and let the moving observer ride in a train whose origin moves with constant velocity v. In the stationary frame the moving origin has the worldline

$$x = vt. \tag{1.1}$$

The next point of the moving lattice traces a parallel line shifted by a fixed amount, then the next one, and so on. In a spacetime plot the moving family of constant-position lines is tilted.

That tilt is the whole content of Galilean kinematics. An event with stationary coordinates (x, t) has, for the moving observer, the same time but a different spatial coordinate, because the moving

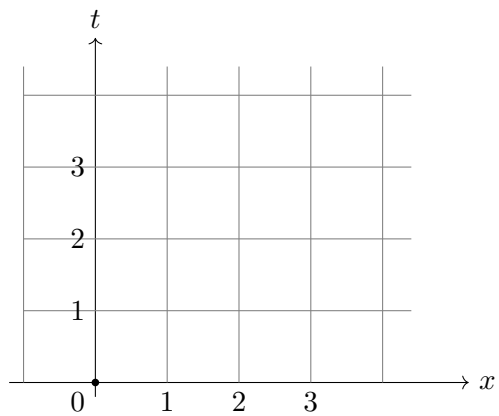


Figure 1.2: A clean reconstruction of the stationary spacetime lattice: constant-position worldlines are vertical and constant-time slices are horizontal.

origin has shifted by the amount vt. Reading directly from the diagram, one arrives at

$$t' = t, \tag{1.2}$$
$$x' = x - vt. \tag{1.3}$$

Susskind is careful about the order here. He does not begin from algebra and then draw a picture afterward. He draws the picture and then reads the equations off the board.

We can invert the relation as well. Since $t = t'$, the inverse form is

$$t = t', \tag{1.4}$$
$$x = x' + vt'. \tag{1.5}$$

The lower inverse equation is cropped in the board frame, and Susskind briefly phrases it as $x = x' + vt$. But because $t = t'$ in Galilean relativity, the intended inverse equation is the canonical one just written.

This is also the point in the lecture where the train analogy and the mathematics meet cleanly. The first pair of equations tells the stationary observer how to convert his description into the moving observer's description; the second pair tells us how to go back. Galilean relativity is, in that sense, a theory of transformations between frames.

1.4 What Galileo Gets Right

Susskind now pauses and asks the right question: what is invariant? He does not want the transformation law to float free of physics.

Take two events at the same time t, with positions x_1 and x_2. In the moving frame,

$$x_1' = x_1 - vt, \tag{1.6}$$
$$x_2' = x_2 - vt. \tag{1.7}$$

Subtracting the two,

$$x_2' - x_1' = (x_2 - vt) - (x_1 - vt) = x_2 - x_1. \tag{1.8}$$

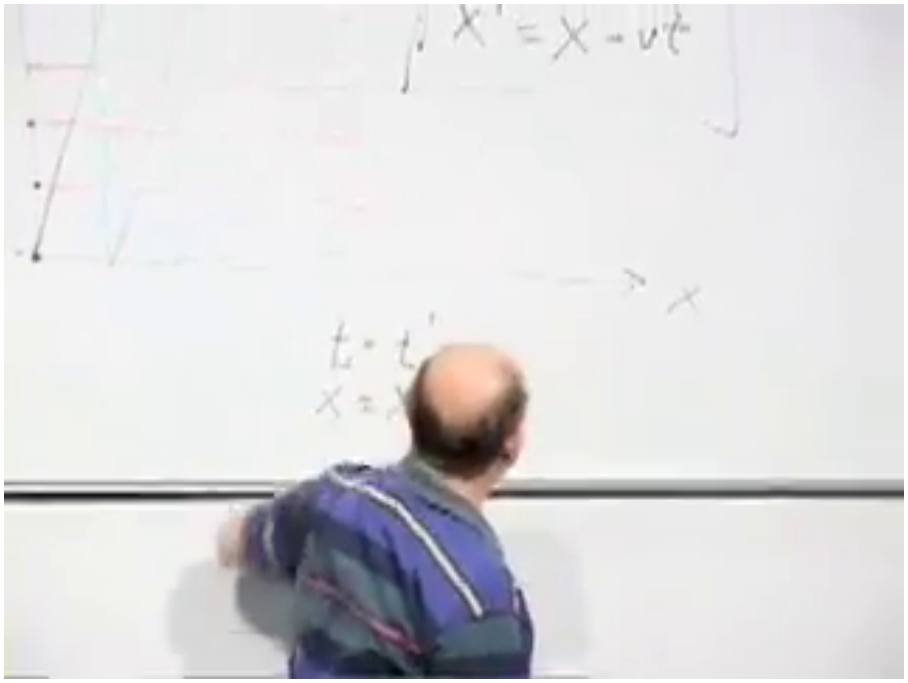


Figure 1.3: Galilean coordinate shift beside the spacetime sketch.

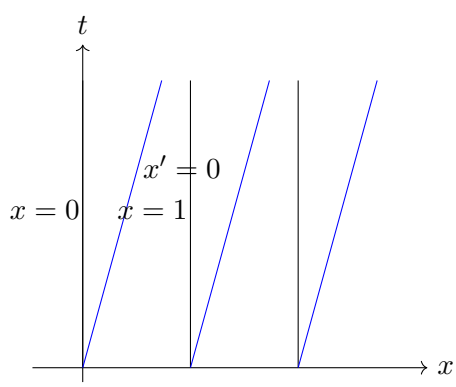


Figure 1.4: Stationary worldlines and the tilted moving family from which the Galilean shift is read.

So the spatial separation of two simultaneous events is invariant under Galilean transformations. The individual coordinates are not invariant, but their difference is.

At this stage the lecture also raises a natural worry. How do we know that time really remains universal when a clock is carried around? Galileo himself did not know in any strict modern sense. It was, as Susskind says, frankly a guess, though a very good one. For ordinary clocks moved around at ordinary speeds, the agreement is fantastically precise. Only at sufficiently high speeds does the guess fail, and that failure is exactly what special relativity is about.

Now let a particle follow a trajectory $x(t)$. The moving observer sees

$$x'(t) = x(t) - vt. \tag{1.9}$$

So position is not invariant. Different observers attach different spatial coordinates to the same particle. Differentiating once,

$$\frac{dx'}{dt} = \frac{dx}{dt} - v, \tag{1.10}$$

and therefore velocity is not invariant either. This is obvious from ordinary experience: if I move along beside your arrow, I judge its speed differently from you.

Differentiate once more:

$$\frac{d^2x'}{dt^2} = \frac{d^2x}{dt^2}, \tag{1.11}$$

because the relative velocity v between inertial frames is constant. Acceleration is invariant.

This is the Newtonian payoff. If the force between two objects depends only on the distance between them, and if that distance is invariant, then the force is also invariant. If acceleration is invariant as well, then

$$F = ma \tag{1.12}$$

has precisely the right structure to serve as a law of motion that all inertial observers can share, provided mass itself is invariant. In this sense Newton's laws are invariant under Galilean coordinate transformations.

The lecture's point is not merely historical courtesy. Galileo and Newton were not making a naive mistake. For mechanics at low velocities, the whole scheme is extraordinarily successful.

1.5 Maxwell's Challenge and Einstein's Return to First Principles

The next move is the crisis. Galilean relativity works beautifully for mechanics, but electromagnetism changes the stakes. Maxwell's equations predict a definite speed for light, and if Maxwell's equations are taken seriously as laws of nature, then the speed of light ought to be the same for every inertial observer.

Galilean kinematics says otherwise. If a light ray in the stationary frame satisfies

$$x = ct, \tag{1.13}$$

then the moving observer would assign

$$x' = x - vt = (c - v)t. \tag{1.14}$$

So under a Galilean transformation the moving observer sees light propagate at speed $c - v$, not at c. That is exactly what one would expect for sound waves or water waves. Indeed Susskind stresses that this was not a crazy expectation. It was the natural expectation.

The historical temptation, before Einstein, was to conclude that there must be a preferred frame, a kind of God's frame, in which Maxwell's equations are exactly right and in which light really travels at speed c. Einstein's move is more radical and more economical. He keeps the relativity principle and forces space and time to adapt. In the lecture this comes in the form of two postulates:

1. The laws of physics are identical in every inertial frame.
2. The speed of light is the same in every inertial frame.

But these postulates do not yet tell us how to assign coordinates. So Einstein goes back to first principles and asks what it means for clocks at different places to be synchronized. The answer is operational. Put an observer at the midpoint between two endpoint clocks. Emit flashes from the endpoints when the endpoint clocks read the same time. If the midpoint observer receives the flashes simultaneously, the clocks are synchronized.

This is the point where the lecture stops being Newtonian in spirit. If a second observer moves past that synchronized apparatus, the two flashes no longer arrive at the moving midpoint at the same instant. The moving observer must therefore conclude that the original emissions were not simultaneous. Simultaneity is not absolute.

1.5.1 Question & Answer

Question. Why not synchronize clocks with sound instead of light?

Answer. One can synchronize clocks with sound in a frame in which the air is at rest. But then one immediately discovers that moving observers measure different sound speeds. Sound propagates in a medium, and motion relative to that medium matters. In an open-air train moving through still air, sound would be a bad messenger of relativity. Light is chosen for the opposite reason: Einstein's postulate is that light has the same speed for every inertial observer, so light is exactly the signal with which the notion of simultaneity must be rebuilt.

The lecture now asks its central quantitative question: if the stationary observer calls one pair of distant events simultaneous, which pair does the moving observer call simultaneous instead?

1.6 The Geometry of Relativity of Simultaneity

Susskind now slows down and makes the question geometric. He sets $c = 1$, so that space and time are measured in compatible units and light rays run at 45°. A right-moving light ray has the form

$$x = t + d, \tag{1.15}$$

and a left-moving light ray has the form

$$x = -t + d'. \tag{1.16}$$

In particular, the light ray from the origin satisfies $x = t$.

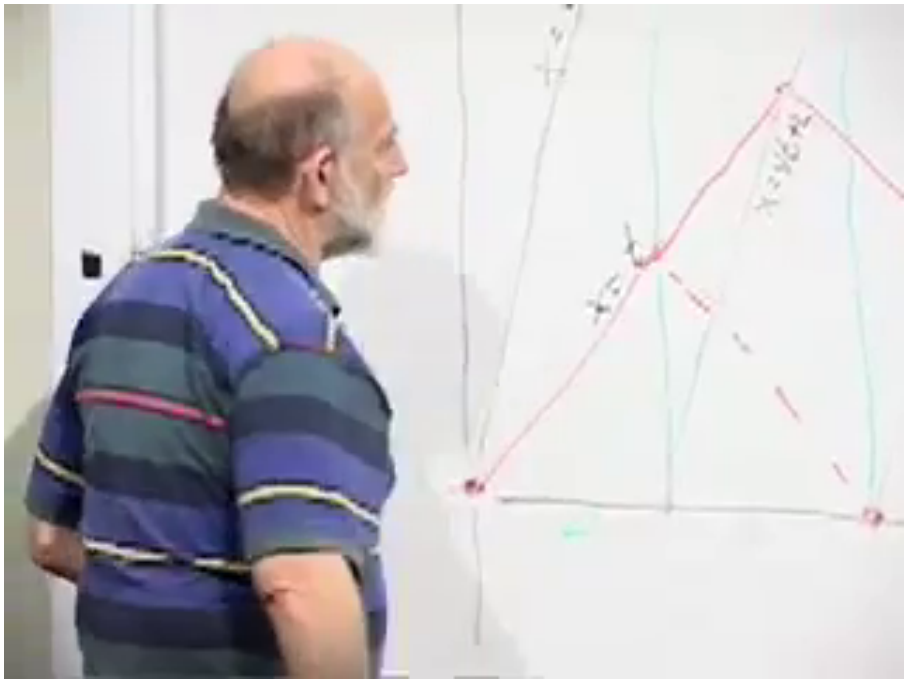


Figure 1.5: Light-ray line $x = t$ in the simultaneity construction.

He then lays out the rest-frame synchronization experiment with endpoints separated by equal distances L. The midpoint observer sits between them. Once the moving frame is added, the question becomes concrete: which event on the right-moving family is simultaneous, in the moving sense, with the event at the origin?

On the board only the label $x = t$ is clearly legible. The shifted moving worldlines are reconstructed from the transcript and from the geometry as

$$x = vt, \qquad x = vt + L, \qquad x = vt + 2L. \tag{1.17}$$

One transcript line later drops the factor of t, but the construction itself makes the intended form unambiguous.

The lecture's rhythm at this point is exact: it first argues qualitatively that the two frames will disagree, and then asks, "Can we do it quantitatively?" The answer is yes.

Worked derivation. Let A be the point where the forward light ray $x = t$ meets the shifted worldline $x = vt + L$. Then A is determined by

$$x = t, \tag{1.18}$$

$$x = vt + L. \tag{1.19}$$

Substituting $x = t$ into the second equation gives

$$t = vt + L, \tag{1.20}$$

so

$$(1 - v)t = L, \tag{1.21}$$

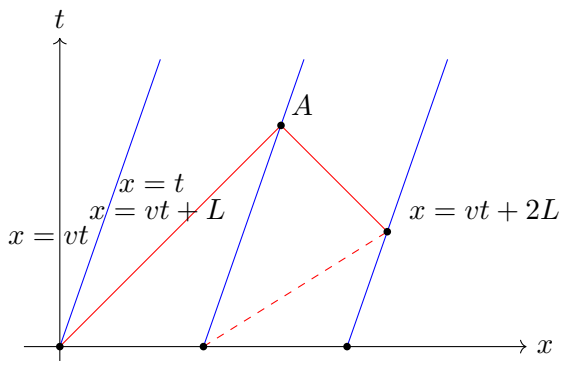


Figure 1.6: Reconstruction of the simultaneity argument. The screenshot preserves the board evidence; the redraw makes the equations explicit.

and therefore

$$t_A = \frac{L}{1-v}, \qquad x_A = \frac{L}{1-v}. \tag{1.22}$$

Now follow a backward-going light ray from A. A left-moving light ray has the form

$$x = -t + b, \tag{1.23}$$

or equivalently

$$x + t = b. \tag{1.24}$$

To determine b, use the fact that the ray passes through A. At A, we have $x = t = L/(1-v)$, so

$$b = x_A + t_A = \frac{2L}{1-v}. \tag{1.25}$$

Thus the backward light ray is

$$x + t = \frac{2L}{1-v}. \tag{1.26}$$

The event we want is where this backward light ray meets the next shifted worldline $x = vt + 2L$. So we solve

$$x + t = \frac{2L}{1-v}, \tag{1.27}$$

$$x = vt + 2L. \tag{1.28}$$

Substituting the second equation into the first yields

$$vt + 2L + t = \frac{2L}{1-v}. \tag{1.29}$$

Rearranging,

$$(1+v)t = 2L\left(\frac{1}{1-v} - 1\right) = 2L\frac{v}{1-v}, \tag{1.30}$$

and therefore

$$t = \frac{2Lv}{1-v^2}. \tag{1.31}$$

Putting this back into $x = vt + 2L$, we obtain

$$x = \frac{2L}{1 - v^2}. \tag{1.32}$$

What matters is not these coordinates by themselves, but their ratio:

$$\frac{t}{x} = v. \tag{1.33}$$

So the moving observer's simultaneity line through the origin is

$$t = vx. \tag{1.34}$$

This is the lecture's geometric payoff. The moving observer's time axis is

$$x = vt, \tag{1.35}$$

because that is the worldline of the moving origin. The moving observer's line of simultaneity is

$$t = vx, \tag{1.36}$$

because those are the points assigned the same moving time. The two are tilted symmetrically with respect to the light rays.

Restoring dimensions requires only one correction:

$$t = \frac{v}{c^2}x. \tag{1.37}$$

This formula makes clear why simultaneity appears absolute in ordinary life. The denominator c^2 is enormous. Only when velocities become appreciable compared with the speed of light does the tilt become visible.

1.7 From Tilted Axes to the Lorentz Transformation

The lecture now performs an instructive cleanup. It erases the clutter of the construction and keeps only what the construction has taught us: the unprimed axes, the primed axes, and the light rays. In units with $c = 1$, the picture has what Susskind calls a lovely symmetry.

The lecture later shifts on the board to upper-case X, T, but the substance is unchanged. We keep lower-case x, t throughout.

The geometry already tells us the zero-sets of the primed coordinates. Since the moving origin is the line $x = vt$, the quantity x' must vanish there. That is the surviving content of Galileo's old formula $x' = x - vt$: even if the exact coefficient changes, the zero-set must remain the same. This motivates

$$x' = \frac{x - vt}{F(v)}. \tag{1.38}$$

Likewise, because the moving simultaneity axis is the line $t = vx$, the quantity t' must vanish there. So we write

$$t' = \frac{t - vx}{G(v)}. \tag{1.39}$$

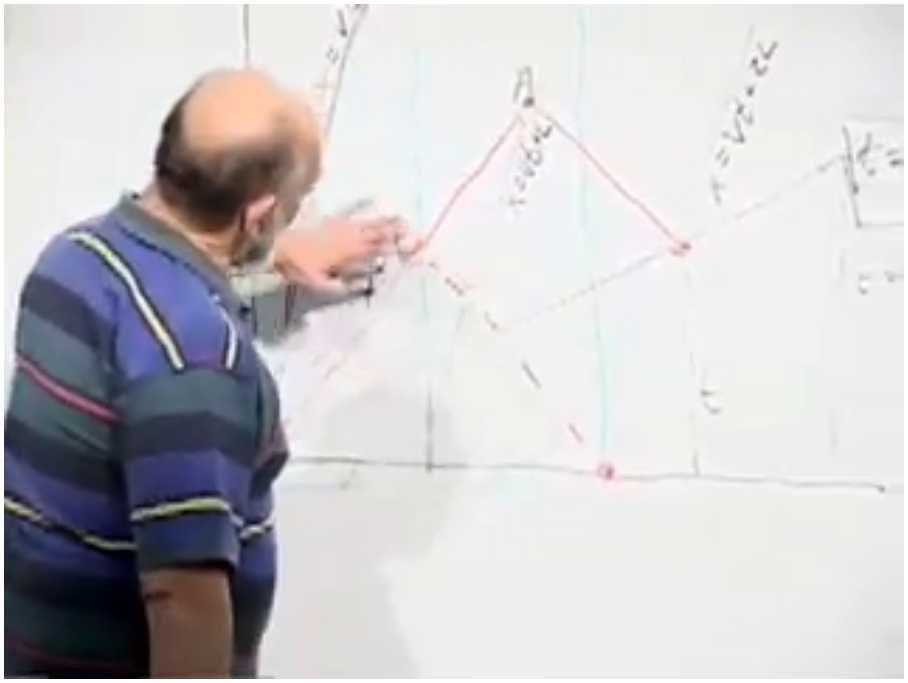


Figure 1.7: Symmetric $c = 1$ spacetime diagram.

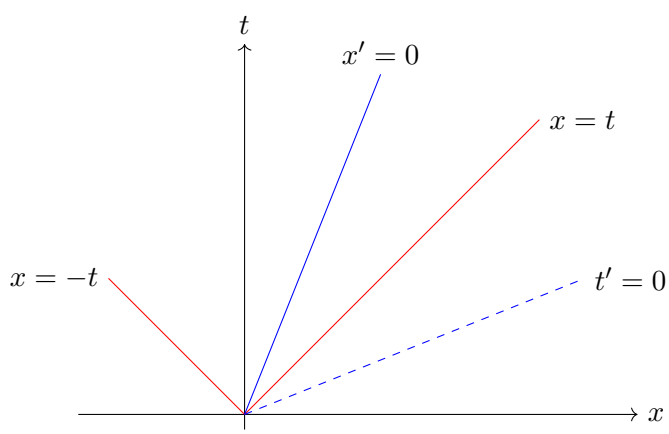


Figure 1.8: Cleaned $c = 1$ diagram showing the primed time axis, the primed simultaneity axis, and the two light rays.

Here $F(v)$ and $G(v)$ are unknown denominator functions. This matters: when the lecture later concludes that $F(v) = \sqrt{1-v^2}$, it is the denominator that equals that square root.

Now use the invariance of light speed. A light ray satisfying $x = t$ in the unprimed frame must satisfy $x' = t'$ in the primed frame. Substituting $x = t$ gives

$$x' = \frac{(1-v)t}{F(v)}, \tag{1.40}$$

$$t' = \frac{(1-v)t}{G(v)}. \tag{1.41}$$

These must agree for every such light ray, so

$$F(v) = G(v). \tag{1.42}$$

Hence

$$x' = \frac{x - vt}{F(v)}, \qquad t' = \frac{t - vx}{F(v)}. \tag{1.43}$$

We still do not know $F(v)$. To determine it, multiply through:

$$F(v)x' = x - vt, \tag{1.44}$$

$$F(v)t' = t - vx. \tag{1.45}$$

To solve for x, multiply the second equation by v and add it to the first. The vt terms cancel:

$$(1-v^2)x = F(v)(x' + vt'), \tag{1.46}$$

so

$$x = \frac{F(v)(x' + vt')}{1-v^2}. \tag{1.47}$$

By the same algebra,

$$t = \frac{F(v)(t' + vx')}{1-v^2}. \tag{1.48}$$

Now comes what Susskind calls the other half of the key. The forward and inverse transformations should have the same form, except that the sign of the relative velocity reverses. This reciprocity condition demands

$$\frac{F(v)}{1-v^2} = \frac{1}{F(v)}. \tag{1.49}$$

Multiplying through by $F(v)$, we obtain

$$F(v)^2 = 1 - v^2, \tag{1.50}$$

and therefore

$$F(v) = \sqrt{1-v^2}. \tag{1.51}$$

The Lorentz transformation in units with $c = 1$ is now fixed:

$$x' = \frac{x - vt}{\sqrt{1-v^2}}, \tag{1.52}$$

$$t' = \frac{t - vx}{\sqrt{1-v^2}}. \tag{1.53}$$

By reciprocity, the inverse relations are

$$x = \frac{x' + vt'}{\sqrt{1 - v^2}}, \tag{1.54}$$

$$t = \frac{t' + vx'}{\sqrt{1 - v^2}}. \tag{1.55}$$

To restore dimensions, we replace v^2 by v^2/c^2, and in the time equation we divide the mixed term by c^2:

$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}}, \tag{1.56}$$

$$t' = \frac{t - (v/c^2)x}{\sqrt{1 - v^2/c^2}}. \tag{1.57}$$

If one later introduces

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}, \tag{1.58}$$

these become the familiar textbook formulas. But the lecture reaches them first in denominator form, and it is worth keeping that sequence.

A direct check. The lecture closes by checking the defining property of the transformation. If

$$x = ct, \tag{1.59}$$

then

$$x' = \frac{(c - v)t}{\sqrt{1 - v^2/c^2}}, \tag{1.60}$$

$$t' = \frac{(1 - v/c)t}{\sqrt{1 - v^2/c^2}}. \tag{1.61}$$

Hence

$$\frac{x'}{t'} = \frac{c - v}{1 - v/c} = c. \tag{1.62}$$

So a right-moving light ray still travels with speed c in the primed frame. The same computation shows that a left-moving light ray satisfying $x = -ct$ becomes $x' = -ct'$.

The lecture also insists on the low-velocity limit. When $v \ll c$, the factors v^2/c^2 and $(v/c^2)x$ are negligible in ordinary circumstances, and the Lorentz transformation reduces to

$$t' \approx t, \qquad x' \approx x - vt. \tag{1.63}$$

Galileo is recovered as an approximation. But the new thing, the genuinely bizarre thing in Susskind's phrasing, is already visible: the surfaces $t =$ constant and $t' =$ constant are different surfaces. Simultaneity is frame-dependent.

1.8 Summary

This lecture unfolds in a deliberate order. We begin with a reference frame understood operationally as a lattice of meter sticks and synchronized clocks. We draw spacetime before we write equations. From that geometry we read the Galilean transformation and then test it: simultaneous distance is invariant, position and velocity are not, acceleration is, and so Newtonian mechanics acquires the right invariant form.

Only after Galileo has been allowed to look strong does the real problem appear. Maxwell and light do not fit Galilean kinematics. Einstein's response is not to abandon relativity, but to return to first principles and rebuild synchronization with light signals. From that reconstruction comes the relativity of simultaneity.

The concrete light-ray construction then gives the moving time axis $x = vt$ and the moving simultaneity axis $t = vx$ in units with $c = 1$. From there, plus the invariance of light speed and the reciprocity of the two frames, the Lorentz transformation follows. The lecture stops before length contraction and time dilation are derived. Its real achievement is more basic: it has shown us that time is not simply time, and that what counts as synchronous in one frame is not what counts as synchronous in another.

Chapter 2

Proper Time, Time Dilation, and Length Contraction

This lecture, though delivered inside the entanglement track, is really a special-relativity lecture. These notes follow Leonard Susskind's order of attack as preserved in the LazyingArt LLC course curation: start with Galileo, correct the kinematics so that light has the same speed for everybody, then use the resulting spacetime geometry to understand proper time, time dilation, the twin paradox, signal exchange, and finally length contraction. The argument is deliberately picture-first. We are not handed isolated formulas; we are shown why each one appears when it does.

2.1 Review: Galilean Relativity and Its Spacetime Grid

Susskind begins by fixing a practical convention. Sometimes the speed of light will be written explicitly as c, and sometimes it will be set equal to 1. We will follow that convention. Most of the algebra is cleaner in units with $c = 1$, and whenever dimensions matter we can restore the missing factors of c by dimensional analysis.

The pre-Einsteinian starting point is Galilean relativity. For motion along a single spatial direction,

$$x' = x - vt, \qquad t' = t. \tag{2.1}$$

This is Newton's world. The moving origin drifts through space, but time is universal. If we solve for the unprimed coordinates, we find

$$x = x' + vt = x' + vt', \qquad t = t'. \tag{2.2}$$

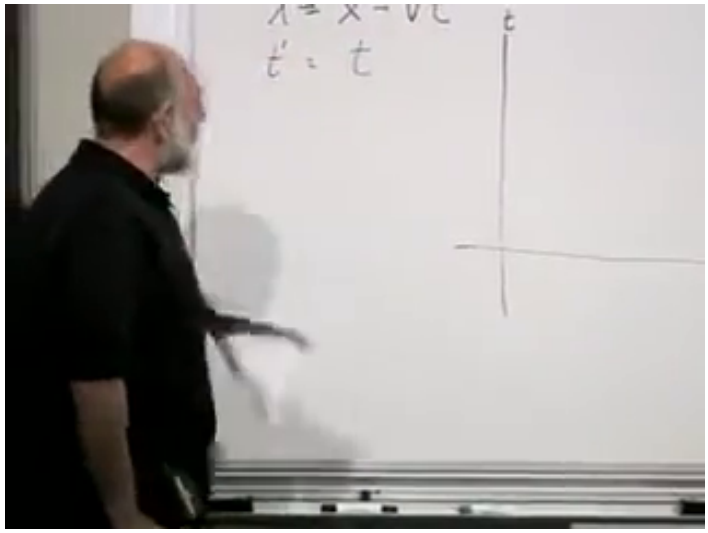
The structure is reciprocal: going from one observer to the other only reverses the sign of the velocity.

Susskind phrases the story in train-and-station language. He imagines himself on the moving train and the audience at rest on the platform, while also reminding us that the whole point of relativity is that either side may regard itself as the one at rest. But before Einstein is allowed to enter, one wants to see exactly what Galileo would have drawn.

The first geometric fact is that the moving observer's spatial origin is the locus $x' = 0$. In the unprimed coordinates this becomes

$$x = vt \iff x' = 0. \tag{2.3}$$

That line is the t'-axis, drawn on the same (x, t) graph as the original axes. The line tilts because the moving origin drifts, but the constant-time slices do not. Since $t' = t$, a surface of constant t' is also a surface of constant t. In Galilean relativity all observers share the same simultaneity.



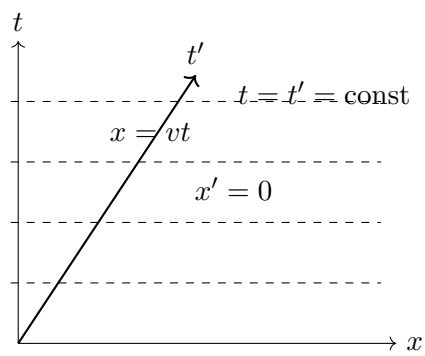


Figure 2.1: The board at the start of the Galilean review, followed by a clean reconstruction of the spacetime picture described immediately afterward in the lecture. The moving origin traces the slanted line $x = vt$, but the constant-time slices remain horizontal because $t' = t$.

This is the baseline. Einstein's correction will be intelligible only if we keep this old grid in view.

2.2 From Galileo to Lorentz

The reason the Galilean formulas fail is equally simple and equally profound: they are inconsistent with the statement that the speed of light is the same in every inertial frame. The corrected formulas are the Lorentz transformations. In units with $c = 1$,

$$x' = \frac{x - vt}{\sqrt{1 - v^2}}, \qquad t' = \frac{t - vx}{\sqrt{1 - v^2}}. \tag{2.4}$$

Restoring dimensions gives

$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}}, \qquad t' = \frac{t - vx/c^2}{\sqrt{1 - v^2/c^2}}. \tag{2.5}$$

Susskind makes two immediate points. First, for everyday velocities the correction is negligible:

$$\sqrt{1-\frac{v^2}{c^2}} \approx 1 \qquad \text{when} \qquad v \ll c. \tag{2.6}$$

So Newtonian physics survives as the low-velocity approximation. Special relativity is not a repudiation of Galileo; it is the refinement that becomes necessary when v approaches c.

Second, the really new feature is not the denominator in x' but the fact that t' is no longer equal to t. Newton believed in a universal time shared by everyone. Einstein's formula says that once the speed of light is universal, time itself must mix with space.

There is also a directional asymmetry built into the formula. If the relative motion is along the x-axis, then the transverse coordinates are unchanged:

$$y' = y, \qquad z' = z. \tag{2.7}$$

Only the coordinate along the direction of motion gets entangled with time.

Susskind also pauses on the denominator. When v approaches c, the factor $\sqrt{1-v^2/c^2}$ tends to zero. The algebra is warning us that one inertial frame cannot move at exactly the speed of light relative to another. Long before one reaches that point, however, measurable deviations from Galilean intuition begin to appear.

2.2.1 Question & Answer

Question. Why do y and z stay unchanged while x mixes with t? And what is the real invariant content of the transverse meter-stick argument?

Answer. The lecture's answer is operational and symmetrical. Imagine two meter sticks oriented transversely to the motion, one carried on the train and one at rest on the platform. Consider an observer halfway between the two frames, a bicyclist in Susskind's language, who sees them approaching with equal and opposite velocities. In that halfway frame the setup is completely symmetric. There is no reason for one transverse rod to pass the other looking shorter. So the transverse directions cannot contract.

But the deeper point is about events. When two endpoints meet, that meeting is a spacetime coincidence. One may mark it by a flash of light, but the flash is only evidence; the coincidence itself is the event. If two events occur at the same spacetime point in one frame, they occur at the same spacetime point in every frame. Thus the transverse rod argument is not merely a visual story. It rests on the invariance of events.

2.3 Light-Cone Coordinates and the Geometry of Lorentz Transformations

At this point Susskind deliberately changes the way he looks at the formulas. He says, in effect: let me show you another way to write the Lorentz transformation. First he gets rid of the c's, because they are a nuisance for the algebra, and then he adds and subtracts the two equations.

Adding the transformed coordinates gives

$$t' + x' = \frac{(t - vx) + (x - vt)}{\sqrt{1 - v^2}} \tag{2.8}$$

$$= \frac{(t + x)(1 - v)}{\sqrt{1 - v^2}} \tag{2.9}$$

$$= (t + x)\sqrt{\frac{1 - v}{1 + v}}. \tag{2.10}$$

Subtracting gives

$$t' - x' = \frac{(t - vx) - (x - vt)}{\sqrt{1 - v^2}} \tag{2.11}$$

$$= \frac{(t - x)(1 + v)}{\sqrt{1 - v^2}} \tag{2.12}$$

$$= (t - x)\sqrt{\frac{1 + v}{1 - v}}. \tag{2.13}$$

This is not merely algebraic housekeeping. The combinations

$$t + x, \qquad t - x \tag{2.14}$$

are coordinates along the two 45° null directions in a spacetime diagram with $c = 1$. In other words, they are adapted to the light cone itself. Under a Lorentz transformation, each of these null coordinates is simply rescaled, and the two scale factors are inverses of one another.

That is the geometric content Susskind wants us to see. A Lorentz transformation is a squeeze: one lightlike direction is stretched while the other is compressed by the reciprocal amount. The crucial thing is what does *not* change. The 45° directions remain 45°. Light rays stay light rays. That is precisely the geometric meaning of the principle that the speed of light is frame-independent.

Once one sees the Lorentz transformation this way, the next question presents itself almost automatically. If the transformation squeezes spacetime but leaves the light cone intact, what quantity does it preserve?

2.4 Proper Time as the Invariant Interval

Before he writes down the formula, Susskind does something wise. He explains what kind of object proper time is supposed to be. Imagine a collection of stationary observers, each carrying a synchronized clock. Their network defines a coordinate time. Now imagine one observer moving through that network with a wristwatch. The wristwatch does not carry the whole stationary grid along with it; it carries only its own ticking. Proper time will be the time read by that moving clock.

With that operational picture in the background, Susskind turns to the geometric analogue of Euclidean distance. For ordinary rotations in the plane, the invariant quantity is

$$s^2 = x^2 + y^2 = x'^2 + y'^2. \tag{2.15}$$

Under a Lorentz transformation, the corresponding invariant is different by a minus sign:

$$t'^2 - x'^2 = t^2 - x^2. \tag{2.16}$$

Proposition 2.1. *Under the Lorentz transformation in $1+1$ dimensions, the quantity $t^2 - x^2$ is invariant.*

Proof. Starting from

$$x' = \frac{x - vt}{\sqrt{1-v^2}}, \qquad t' = \frac{t - vx}{\sqrt{1-v^2}}, \tag{2.17}$$

we compute

$$\begin{aligned} t'^2 - x'^2 &= \frac{(t-vx)^2 - (x-vt)^2}{1-v^2} && (2.18)\\ &= \frac{t^2 - 2tvx + v^2x^2 - x^2 + 2xvt - v^2t^2}{1-v^2} && (2.19)\\ &= \frac{(1-v^2)t^2 - (1-v^2)x^2}{1-v^2} && (2.20)\\ &= t^2 - x^2. && (2.21) \end{aligned}$$

The cross terms cancel, and the common factor $1 - v^2$ drops out. □

In three spatial dimensions the corresponding statement is

$$t'^2 - x'^2 - y'^2 - z'^2 = t^2 - x^2 - y^2 - z^2. \tag{2.22}$$

The lecture mostly works along one axis, but this fuller expression makes clear that the invariant does not care which spatial direction we happened to choose for the relative motion.

Definition 2.2. If one event is the origin and a second event lies in the time-like region, its proper time from the origin is

$$\tau = \sqrt{t^2 - x^2} \tag{2.23}$$

in units with $c = 1$. More generally, one applies the same invariant to coordinate differences between two events:

$$\Delta\tau^2 = \Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2. \tag{2.24}$$

Why the square root? Because the analogy is with ordinary distance. In Euclidean geometry the distance between two points is the square root of the invariant quadratic form. Here the invariant quadratic form has a minus sign in the spatial directions, but the logic is the same.

Now the sign difference begins to matter. If $t^2 > x^2$, the quantity under the square root is positive and the proper time is real. If $t^2 = x^2$, the proper time is zero. If $t^2 < x^2$, the separation is space-like, and in the lecture Susskind says bluntly that the proper time is imaginary. The polished statement is that space-like separated points are not related by ordinary elapsed clock time, but his warning is useful: Minkowski geometry is not Euclidean geometry with a new accent; the minus sign changes everything.

The operational interpretation now falls into place. For a clock at rest in the unprimed frame, $x = 0$, so

$$\tau = \sqrt{t^2} = t. \tag{2.25}$$

Thus proper time agrees with ordinary time for an observer at rest.

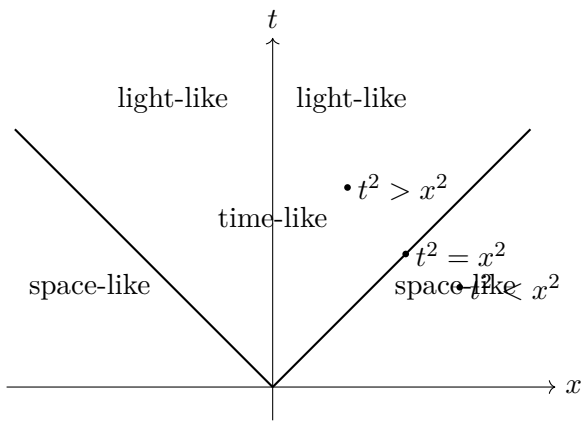


Figure 2.2: The light cone in $1+1$ dimensions. Inside the cone the interval is time-like and the proper time is real; on the cone the proper time is zero; outside the cone the separation is space-like.

For a moving clock, we may always transform to the frame in which the clock is momentarily at rest. Along that worldline $x' = 0$, and so

$$\tau = \sqrt{t'^2 - x'^2} = t'. \tag{2.26}$$

This is the heart of the lecture's definition: moving clocks tick off proper time. The invariant interval is not just a formula that everyone agrees on; it is the quantity that the traveling wristwatch actually reads.

2.5 Time Dilation and Relativity of Simultaneity

Now Susskind uses proper time as a tool. Consider the simplest possible moving worldline: the origin of the primed frame itself. In the unprimed coordinates this worldline is

$$x = vt \iff x' = 0. \tag{2.27}$$

So the slanted line in the spacetime diagram is not merely a particle trajectory. It is the moving observer's time axis.

To find the time read by the moving clock, use the invariant interval:

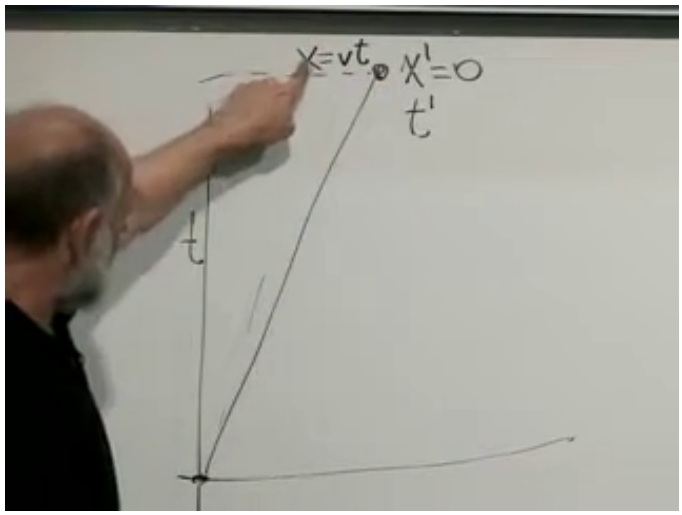
$$t'^2 - x'^2 = t^2 - x^2. \tag{2.28}$$

On the moving origin, $x' = 0$, while in the stationary frame $x = vt$. Therefore

$$t'^2 = t^2 - x^2 \tag{2.29}$$
$$= t^2 - v^2 t^2 \tag{2.30}$$
$$= (1 - v^2)t^2. \tag{2.31}$$

Taking the positive square root, we obtain

$$t' = t\sqrt{1 - v^2}. \tag{2.32}$$



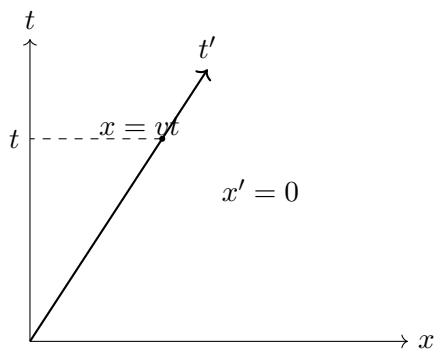


Figure 2.3: The key board frame for the moving-origin construction, paired with a clean reconstruction. The slanted line is identified on the board with both $x = vt$ and $x' = 0$; in the primed frame it is the t'-axis.

With the speed of light restored,

$$t' = t\sqrt{1 - \frac{v^2}{c^2}}. \tag{2.33}$$

This is the famous time-dilation formula. Between the same two spacetime events, the moving clock records less elapsed time than the synchronized stationary grid. In the stationary observer's description, the moving clock runs slow.

The lecture is slightly ragged in speech at this point, but the invariant interval leaves no ambiguity. Once one has written $t'^2 = (1 - v^2)t^2$, the result is fixed.

2.5.1 Question & Answer

Question. How can each observer say that the other clock runs slow? That sounds contradictory.

Answer. The contradiction is only apparent, because the two observers are not comparing the same pairs of events. The stationary observer uses surfaces $t = \text{const}$. The moving observer uses

surfaces $t' = \text{const}$. From

$$t' = \frac{t - vx}{\sqrt{1 - v^2}}, \tag{2.34}$$

we see that holding t' fixed means

$$t - vx = \text{const}. \tag{2.35}$$

So the moving observer's simultaneity slices are tilted relative to the stationary observer's.

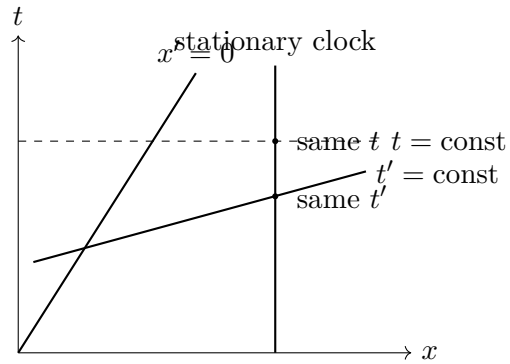


Figure 2.4: Relativity of simultaneity. The stationary observer compares clock readings on a horizontal slice $t = \text{const}$, while the moving observer compares them on a tilted slice $t' = \text{const}$. The two statements about "the other clock" therefore refer to different event pairs.

This is the lecture's resolution of the reciprocity puzzle. Moving clocks appear slow to stationary observers, and stationary clocks appear slow to moving observers, because the very notion of "at the same time" is frame-dependent.

2.6 Twin Paradox, Ideal Clocks, and Accelerated Motion

Having established that clocks measure proper time, Susskind turns to the twin paradox. Two twins are born at the same event. One stays at home on a vertical worldline. The other is shot outward at speed v, turns around, and comes back. When they meet again, they compare everything that can function as a clock: wristwatches, heartbeats, beard growth, metabolic aging. The question is not which clock was more poetic. The question is which worldline accumulated more proper time.

Let the stay-at-home twin measure a time T on the outward leg and another T on the return leg. Then

$$\tau_{\text{home}} = 2T. \tag{2.36}$$

For the traveling twin, the turning point has coordinates

$$x = vT, \qquad t = T. \tag{2.37}$$

Therefore one leg contributes

$$\tau_{\text{leg}}^2 = T^2 - x^2 \tag{2.38}$$

$$= T^2 - v^2T^2 \tag{2.39}$$

$$= (1 - v^2)T^2, \tag{2.40}$$

so

$$\tau_{\text{leg}} = T\sqrt{1 - v^2}. \tag{2.41}$$

The return leg contributes the same amount, hence

$$\tau_{\text{travel}} = 2T\sqrt{1 - v^2}. \tag{2.42}$$

Since $1 - v^2 < 1$, the traveling twin returns younger.

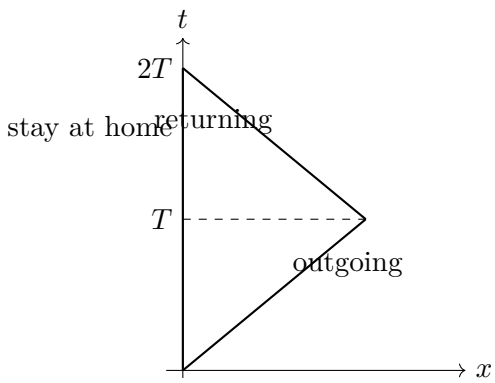


Figure 2.5: Twin paradox as a comparison of proper times along two worldlines: one straight and one broken. The clocks disagree because the paths through spacetime are different, not because the geometry is inconsistent.

Susskind is careful here. He does not leave the result as a mere slogan. The traveling twin is younger for the same reason that a broken path in geometry has a different length from a straight one. Minkowski geometry is unfamiliar, but it is still geometry.

2.6.1 Question & Answer

Question. Does the presence of acceleration destroy the argument? After all, the turnaround is not an inertial segment.

Answer. No one should pretend the acceleration is absent. It is present, and it is one of the things that breaks the symmetry between the twins. But that does not spoil the proper-time calculation.

First, the sharp corner in the spacetime diagram is only an idealization. We may smooth it into a finite turnaround without changing the conclusion.

Second, the lecture introduces an additional physical assumption: the assumption of an ideal clock. A sufficiently small, well-built clock is taken to be insensitive to modest accelerations, so that it continues to tick off proper time. Then a curved worldline can be approximated by many short inertial segments, and one adds the proper times segment by segment:

$$\tau_{\text{path}} \approx \sum_i \sqrt{\Delta t_i^2 - \Delta x_i^2}. \tag{2.43}$$

This is the spacetime analogue of computing the length of an ordinary curve by dividing it into many small straight pieces.

Susskind is explicit that this ideal-clock behavior is a physical assumption, not something proved from the Lorentz algebra alone. Small, faithful clocks may be engineered to withstand the relevant accelerations; big sloppy clocks may not. The geometry tells us what an ideal clock records.

The lecture then pushes the same issue into a more operational form. Instead of comparing final clock readings only, we imagine marking one-second intervals of proper time along both worldlines and sending light pulses back and forth. That way one asks not just what the clocks finally say, but how each observer actually receives the other's signals.

2.7 Signals, Closed Spatial Loops, and Length Contraction

Now the lecture leaves the purely geometrical comparison of two final clock readings and asks what each observer actually sees. Suppose the stay-at-home twin emits one light pulse every second of proper time, and the traveling twin does the same. On the outward leg the received pulses are stretched out; on the inward leg they become bunched together. Susskind explicitly identifies this as a Doppler-type effect, though he does not need the full formalism of relativistic Doppler shift to make the point.

What matters is the rhythm of reception. Early pulses arrive late and sparse; later pulses arrive fast and compressed. No pulses go missing. Each observer eventually receives all the pulses sent by the other, but at rates that reflect the relative motion. This operational discussion is important in the lecture because it keeps proper time from becoming a purely formal invariant. The geometry is read out by actual light signals.

A question about energy is raised at this point and then postponed. Susskind refuses to answer it too early. At this stage the lecture is about coordinates, clocks, and signals, not yet about the full relativistic bookkeeping of energy for accelerated observers.

2.7.1 Question & Answer

Question. What if one spatial direction were compact, so that an observer could move uniformly around a closed loop and return without the usual turn-around? Would the asymmetry survive?

Answer. Here the lecture becomes exploratory and deliberately unfinished. Susskind first says that the asymmetry should survive, because the moving observer would discover a surprise: the same places would recur. But he then pulls back and refuses to give an unthought-out answer. The right way to preserve this beat is not to turn it into a polished theorem. It is to record the qualitative claim and the fact that the full geometry is postponed.

Only after this detour does Susskind return to the next standard effect, and he does so with a piece of methodological advice strong enough to count as doctrine: when you want to understand a special-relativity problem, draw a picture.

Take a meter stick at rest in the unprimed frame. Its left end lies at $x = 0$, its right end at $x = 1$. As time passes, the rod sweeps out a ribbon in spacetime, bounded by two vertical worldlines. The question is: what length does the moving observer assign to that ribbon?

The answer begins with the meaning of measurement. A length in the moving frame is the separation of the rod's endpoints at the same instant of *moving-frame* time. That is the local conceptual

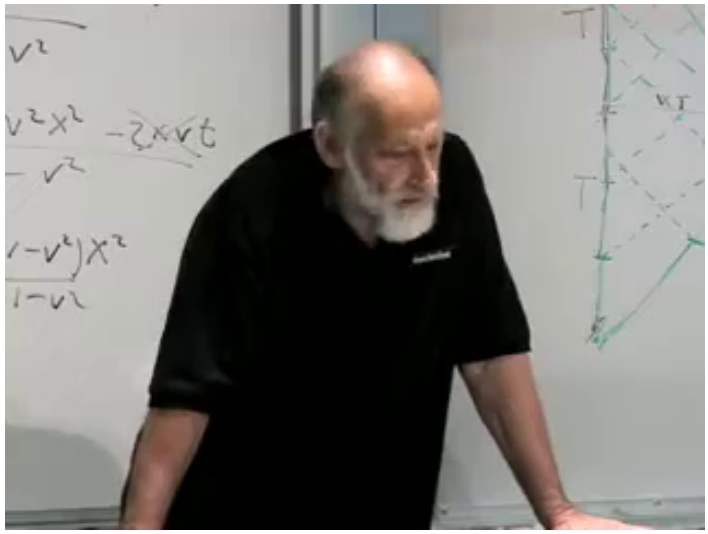


Figure 2.6: The later board during the compact-direction discussion. The left-board algebra is too fragmentary to transcribe safely, but the right-hand geometric sketch shows that the lecture had shifted into an asymmetry puzzle rather than a finished derivation.

obstacle in this part of the lecture. One must not look at arbitrary points on the ribbon; one must intersect the ribbon with a single constant-t' slice.

The clean relation is

$$t - vx = \text{const} \tag{2.44}$$

for constant t' in units with $c = 1$. In particular, the slice through the origin with $t' = 0$ satisfies

$$t = vx. \tag{2.45}$$

The lecture's spoken line wavers momentarily here, but this is the correct standard form. When the slice hits the right endpoint worldline $x = 1$, the corresponding unprimed time is therefore

$$t = v. \tag{2.46}$$

Because the separation from the origin to the selected endpoint is now space-like, it is cleaner to reverse the sign of the invariant interval and speak of the squared proper distance:

$$x'^2 - t'^2 = x^2 - t^2. \tag{2.47}$$

On the chosen slice, $t' = 0$. At the selected endpoint, $x = 1$ and $t = v$. Therefore

$$x'^2 = 1 - v^2. \tag{2.48}$$

Equivalently, in the lecture's board language one may write

$$-x'^2 = v^2 - 1, \tag{2.49}$$

which is the same statement with the signs not yet cleaned up. Either way the conclusion is

$$x' = \sqrt{1 - v^2}. \tag{2.50}$$

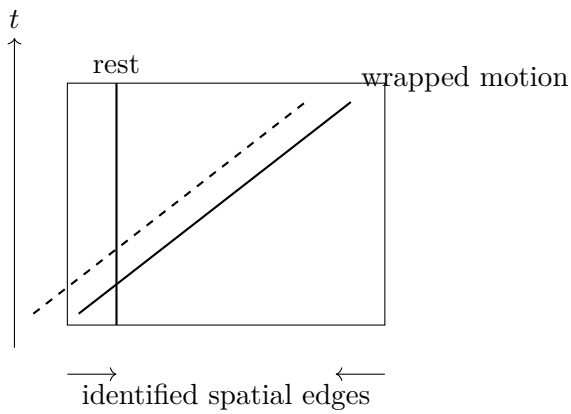


Figure 2.7: A cautious schematic reconstruction of the compact-direction question, drawn on an unwound strip. This is not a completed theorem from the lecture; it is only a clean rendering of the qualitative asymmetry being discussed.

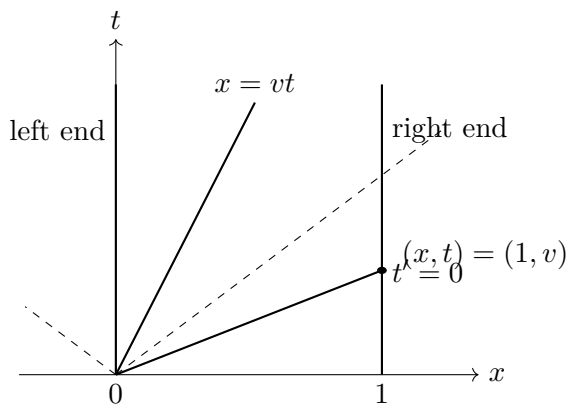


Figure 2.8: Length contraction from the spacetime picture Susskind insists on drawing. The rod is at rest in the unprimed frame, so its endpoints trace a vertical ribbon. The moving observer measures the rod by slicing that ribbon with a constant-t' line.

Thus a rod of rest length 1 is seen in the moving frame to have length

$$L' = \sqrt{1 - v^2}. \tag{2.51}$$

For a general rest length L,

$$L' = L\sqrt{1 - v^2} \tag{2.52}$$

in units with $c = 1$, or

$$L' = L\sqrt{1 - \frac{v^2}{c^2}} \tag{2.53}$$

when c is restored.

This is the companion to time dilation. Time dilation came from following a moving worldline and comparing elapsed proper time. Length contraction comes from slicing a stationary ribbon by the moving observer's simultaneity plane. Both are consequences of the same Lorentz mixing of space and time.

The lecture's final operational payoff is immediate. In the astronaut's frame the distance between distant stars is Lorentz-contracted, so a long trip may take a short proper time without anything moving faster than light. The stars are not seen to outrun light; they are simply seen at a reduced spatial separation in the moving frame.

2.8 Summary

The lecture begins with Galileo on purpose. In the Galilean picture the moving origin tilts, but time slices remain common to everyone. Einstein's correction changes exactly that point: x and t mix, the light cone is preserved, and the invariant quantity is not $x^2 + y^2$ but $t^2 - x^2 - y^2 - z^2$.

From there the lecture unfolds in a strict order. First we reinterpret the Lorentz transformation as a squeeze along null directions. Then we identify the invariant interval. Then we define proper time and recognize it as the time read by a moving clock. Time dilation follows from evaluating that invariant on the worldline $x = vt$. The reciprocity puzzle is resolved by relativity of simultaneity. The twin paradox is resolved by comparing proper times along different paths, with the ideal-clock assumption made explicit. Signal exchange gives the operational picture. A compact-direction asymmetry is raised but deliberately left provisional. Only after that does Susskind return to length contraction, again by drawing the spacetime picture first.

What remains after all the formulas is really a method. Draw the diagram. Identify the invariant. Ask which events are being compared. Then let the geometry tell you what clocks, rods, and signals will do.

Chapter 3

From Heavy-Ion Fireballs to Four-Vectors

These notes follow Leonard Susskind's lecture, with curation by LazyingArt LLC, and keep close to the order in which the lecture actually unfolds. We begin in the middle of a discussion about heavy-ion collisions and a five-dimensional black-hole picture of QCD. Only after that detour has done its motivational work does the lecture pivot back to special relativity. From there the progression is deliberate: a paradox about simultaneity, a cleanup of spacetime notation, rotations and Lorentz transformations in matrix form, the composition of boosts, and finally the relativistic replacement for ordinary velocity, together with the first appearance of proper acceleration and four-momentum.

3.1 Heavy-Ion Collisions, Horizons, and the Fifth-Dimensional Picture

The lecture does not open from a formal beginning. It opens with a question about the complexity of the black hole that is supposed to describe the aftermath of a violent hadronic collision. Susskind answers not with a formula, but with a picture.

We are asked to imagine a fifth dimension extending between a floor and a ceiling. There is a gravitational field in that fifth-dimensional direction, and it pulls excitations downward toward the floor. The excitations may be thought of schematically as little strings. When they settle near the floor, they are hadrons: protons, neutrons, mesons, the large composite objects of QCD. If they remain near the ceiling, they are more elementary excitations: gravitons, photons, and other non-hadronic degrees of freedom. In that simple geometric language, the distinction between composite hadrons and more elementary quanta is encoded in where they live along the extra dimension.

Now collide two hadrons. In the five-dimensional picture this means colliding two objects that live near the floor. They do not produce a small rigid remnant. They produce a large splash of energy, a puddle or lump deposited in the gravitational field. That lump spreads, settles, and is usefully described as a black-hole-like object. Susskind is careful about what kind of black hole this is. It is not an astronomical four-dimensional black hole, nor the Planckian object one might imagine hanging from the ceiling. It is a five-dimensional horizon-bearing lump sitting on the floor. It has a

horizon, temperature, entropy, and the familiar thermodynamic attributes of black holes.

At very high energies the incoming hadrons are Lorentz contracted into thin pancakes. Two such pancakes collide and produce a large hot region, more like a fluid lump than a rigid body. The gravitational field pulls this lump down and spreads it until it comes to something like equilibrium. If that were the whole story, the lump would simply sit there. But it radiates. In this picture the outgoing hadrons are identified with Hawking radiation emitted from the black-hole-like object. The swarm of particles seen in the detector is the field-theoretic image of horizon radiation.

The lecture then pauses over the status of this picture. One can take the fifth dimension literally, or one can regard it as a mathematically equivalent description of strongly coupled QCD, useful precisely because it turns impossible quantum-field-theoretic calculations into more tractable gravitational ones. Susskind leaves the ontological question open, but he does not leave the practical point open: the dual picture captures real features of the quark-gluon plasma, especially its unexpectedly low viscosity, in a way that was not anticipated in the older weak-coupling mindset. In that sense the picture has already earned its keep.

3.2 Collider Energetics and Why Beam Geometry Matters

The discussion then shifts, under pressure from the room, from conceptual interpretation to collider practice. Why collide two beams head-on instead of firing one beam into a stationary target? Why must accelerators be so large? Why does beam geometry matter so much? These questions are not yet the main subject of the lecture, but they motivate the return to relativity by forcing us to think about energy in the center-of-mass frame.

For two highly relativistic particles with equal and opposite momenta p, the lecture-level estimate is that the center-of-mass energy is simply

$$E_{\text{cm}}^{\text{head-on}} \approx 2p \qquad (c = 1). \tag{3.1}$$

Here we are using the relativistic fact that, for very energetic particles, energy and momentum are approximately equal in $c = 1$ units.

If instead a fast beam of momentum p strikes a stationary target of mass m, the center-of-mass energy is much smaller. The lecture gives the rough scaling

$$E_{\text{cm}}^{\text{fixed target}} \sim \sqrt{mp} \qquad (c = 1), \tag{3.2}$$

to emphasize that a tremendous amount of beam energy is wasted when one of the two participants is left at rest. The purpose of the estimate is not to provide a careful invariant-mass derivation, but to make the practical point vivid: a head-on collider is enormously more efficient at concentrating energy into the interaction region.

That is the first engineering lesson. The second concerns size. Electric and magnetic fields can only be made so strong before engineering and material constraints intervene. In a linear accelerator one gives the particle repeated pushes as it moves along. More distance means more pushes and therefore more energy. In that approximate sense the energy of the machine scales with the useful accelerating length.

There is also a second limitation, and the lecture stresses that it is very important. Charged particles moving in circles are accelerated even when their speed is constant, simply because their direction is

changing. The tighter the circle, the larger the acceleration. Large acceleration produces radiation, and radiation means energy loss. If the circular path is too tight, the machine pours energy into the beam only to have the beam radiate it away.

That is why, if the goal is to dump a large amount of energy into a small volume and see what comes out, one would like to accelerate particles as nearly in a straight line as possible. Circular machines survive because one can use the same bending magnets over and over again, repeatedly pushing the beam around the ring; but the price is synchrotron radiation.

The electron–proton contrast fits naturally into this picture. Electrons are much lighter than protons. To carry a given energy they must be pushed much closer to the speed of light, and in a circular machine that means greater acceleration and greater radiative loss. Protons, being much heavier, can circulate with much less energy loss at the same speed and therefore carry much more useful energy in a circular collider. The lecture's point is not to turn this into accelerator design theory, but to clear away the practical questions before making the announced return to relativity.

That announcement is explicit. Having answered the room, Susskind says, in effect: now let us get back to the special theory of relativity.

3.3 Relativity of Simultaneity Through a Spacetime Diagram

The re-entry point is not formalism but paradox. Special relativity, as Susskind presents it here, is best entered by confronting the conceptual mistake that repeatedly causes trouble: we carry into relativity a pre-relativistic idea of simultaneity, and then we are surprised when the answers seem contradictory.

The example is the garage-and-limousine puzzle. A short garage and a long limousine approach one another. In the garage frame the moving limousine is Lorentz contracted, so one expects that it can fit. In the limousine frame the moving garage is Lorentz contracted, so one expects that it cannot fit. The puzzle is designed to force us to ask exactly what we mean when we say that the whole limousine is inside the garage.

3.3.1 Question & Answer

Question. Can the limousine ever fit entirely inside the garage?

Answer. The lecture first makes a practical adjustment: open the back door of the garage. We are not interested in crashing the driver into the rear wall. We want only to know whether there is any instant at which the whole car is inside.

Now draw a spacetime diagram. Time is vertical, space is horizontal, and because we work in one spatial dimension here, x is the only spatial coordinate we need. In units with $c = 1$, light rays travel on lines at 45°, so the light cone gives the geometry its basic orientation. The garage frame has the ordinary t-axis and x-axis. The moving frame has a tilted t'-axis and a tilted x'-axis. Those tilted axes are the visual representation of the fact that the moving observer's notion of simultaneity is different from that of the stationary observer.

The front and back ends of the garage are at fixed positions in the garage frame, so their worldlines are vertical. The front and back ends of the limousine are moving, so their worldlines are slanted

and parallel. The question "Is the whole limousine inside the garage?" means: is there a slice of simultaneity on which both the front and rear ends of the limousine lie between the two garage worldlines?

In the garage frame the answer is yes. There are horizontal lines—constant t slices—for which the back end of the limousine has already entered and the front end has not yet emerged. On such a slice the whole Lorentz-contracted limousine lies inside the garage.

In the limousine frame the answer is no. There, simultaneity is measured along lines parallel to the x'-axis, not along horizontal lines. If we take the event at which the rear end enters and draw the moving observer's simultaneous slice through it, that slice meets the front worldline only after the front has already gone out the back. On the limousine's own slice of simultaneity, the whole car is never inside at once.

There is therefore no contradiction. The hidden word was "simultaneously." Once we distinguish the stationary observer's simultaneity from the moving observer's simultaneity, the paradox evaporates. The lecture then adds an important geometric remark: the segment cut out of the limousine by a horizontal slice is its Lorentz-contracted length in the garage frame, while the proper length of the limousine is measured on a slice simultaneous in the limousine frame. Those are not the same slice. The paradox lives in that difference.

This little puzzle is not peripheral to the chapter. It is the lecture's deliberate way back into relativity. Diagram first, moral second: simultaneity is frame-dependent, and many apparent contradictions are nothing more than a failure to say in which frame the word "simultaneous" is being used.

3.4 Four Coordinates, Covariant Notation, and the Invariant Interval

Only now does the lecture turn to the formal machinery, and even here Susskind insists that most of the mathematics is merely notation. The point of the notation is not abstraction for its own sake. The point is to package the four coordinates in a way that keeps control of the minus signs.

We begin with the four coordinates of spacetime. In the lecture these are lumped together into a single object, with the slightly idiosyncratic convention that the time coordinate is the 0-component even though it is often written last:

$$X^1 = x, \qquad X^2 = y, \qquad X^3 = z, \qquad X^0 = t. \tag{3.3}$$

Thus, schematically,

$$X^\mu = (x, y, z, t), \tag{3.4}$$

with the understanding that the final entry is really X^0.

Susskind then introduces the covariant form of the same object. Nothing new has been added physically. We are describing the same spacetime point, but in a notation that flips the sign of the spatial components:

$$X_\mu = (-x, -y, -z, t). \tag{3.5}$$

This is the lecture's sign convention. In modern language, it is the signature $+---$, but the lecture does not need that phrase. It needs only the operational rule: lower the index and the first three components change sign.

A quick example, very much in the spirit of the lecture, is useful. If a point has coordinates

$$(x, y, z, t) = (3, 2, 4, 10), \tag{3.6}$$

then

$$X^\mu = (3, 2, 4, 10), \qquad X_\mu = (-3, -2, -4, 10). \tag{3.7}$$

This is not a different point. It is the same point written in a notation that is designed to make invariant expressions come out cleanly.

The main invariant then appears immediately:

$$X_\mu X^\mu = t^2 - x^2 - y^2 - z^2. \tag{3.8}$$

This is exactly the proper interval from the origin to the point (x, y, z, t). In differential form,

$$d\tau^2 = dt^2 - dx^2 - dy^2 - dz^2, \tag{3.9}$$

again in $c = 1$ units. Along a timelike worldline, $d\tau$ is the proper time measured by a clock moving with the particle.

At this point Einstein's summation convention becomes useful. Since expressions of the form "sum over one time component with a plus sign and three spatial components with minus signs" occur constantly in relativity, we adopt the rule that whenever an index appears once upstairs and once downstairs, we sum over it. Thus we no longer write the summation sign explicitly; we simply write

$$X_\mu X^\mu, \tag{3.10}$$

and the sum over $\mu = 0, 1, 2, 3$ is understood.

This is the lecture's real point about notation. Relativity generates many expressions of this kind, and without a compact device the algebra becomes cluttered with repeated sums and signs. The notation does not replace the geometry; it stabilizes it.

3.5 Rotations, Matrices, and the Structure of Lorentz Transformations

The lecture now shifts to transformation laws in general. Before mixing time and space, Susskind takes the easier symmetry first: ordinary spatial rotations. This is not a digression. It is a pedagogical ramp. One learns how a transformation acts on components in a familiar case before returning to Lorentz transformations.

Consider a rotation about the z-axis by an angle θ. Then

$$x' = x\cos\theta + y\sin\theta, \tag{3.11}$$

$$y' = -x\sin\theta + y\cos\theta, \tag{3.12}$$

$$z' = z. \tag{3.13}$$

The point of the example is not just the formula itself. It is what the formula preserves. The squared distance from the origin remains unchanged. We can verify this directly:

$$x'^2 + y'^2 = (x\cos\theta + y\sin\theta)^2 + (-x\sin\theta + y\cos\theta)^2 \tag{3.14}$$

$$= x^2(\cos^2\theta + \sin^2\theta) + y^2(\sin^2\theta + \cos^2\theta) + 2xy\sin\theta\cos\theta - 2xy\sin\theta\cos\theta \tag{3.15}$$

$$= x^2 + y^2. \tag{3.16}$$

Since $z' = z$, we conclude that

$$x'^2 + y'^2 + z'^2 = x^2 + y^2 + z^2. \tag{3.17}$$

That is the lecture's template for symmetry: the components change, but a geometrically meaningful quantity remains invariant.

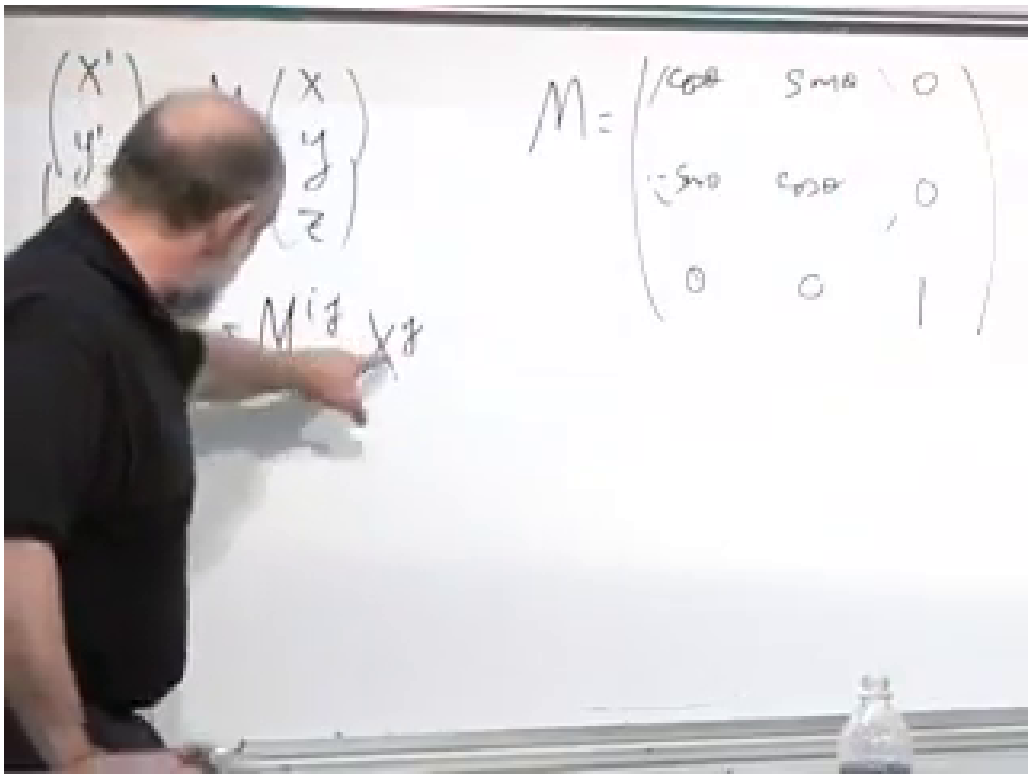


Figure 3.1: Rotation matrix acting on spatial coordinates.

The lecture then rewrites the same transformation in matrix language. That board layout is important and should be preserved. On the left is the vector relation, on the right the explicit matrix:

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = M \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \qquad M = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{3.18}$$

Beneath the vector equation the board shows an indexed form, partly hidden by Susskind's arm. The natural cleaned reconstruction is

$$x'^i = M^i{}_j \, x^j, \tag{3.19}$$

with $i, j = 1, 2, 3$. We should regard this as a cautious normalization of the board line rather than a letter-for-letter transcription.

Once this is in hand, the move to Lorentz transformations becomes natural. A boost along the

x-axis mixes x and t, while leaving y and z untouched. In the lecture's $c = 1$ units,

$$x' = \frac{x - vt}{\sqrt{1 - v^2}}, \tag{3.20}$$

$$y' = y, \tag{3.21}$$

$$z' = z, \tag{3.22}$$

$$t' = \frac{t - vx}{\sqrt{1 - v^2}}. \tag{3.23}$$

It is convenient here to write

$$\gamma = \frac{1}{\sqrt{1 - v^2}} \qquad (c = 1), \tag{3.24}$$

so that the boost takes the compact form

$$x' = \gamma(x - vt), \tag{3.25}$$

$$t' = \gamma(t - vx). \tag{3.26}$$

In matrix form,

$$\begin{pmatrix} x' \\ y' \\ z' \\ t' \end{pmatrix} = \begin{pmatrix} \gamma & 0 & 0 & -\gamma v \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\gamma v & 0 & 0 & \gamma \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix}. \tag{3.27}$$

This is the cleaned standard form of the board calculation. The spoken algebra in the lecture is somewhat messy, but the intended object is clear: a Lorentz transformation is also a matrix acting linearly on a column of components.

At this stage the lecture makes another conceptual point. Rotations and boosts together generate all inertial frames related by uniform motion. If the relative motion is not along one of the original coordinate axes, we first rotate the spatial axes until the motion lies along the new x-direction, then perform the boost there. That is why the rotation section is not ornamental. It is part of the general structure of relativistic symmetry.

The first substantial payoff of the matrix language comes from composing boosts. Suppress the inert y and z directions and write the x-t boost as

$$B(v) = \gamma_v \begin{pmatrix} 1 & -v \\ -v & 1 \end{pmatrix}, \qquad \gamma_v = \frac{1}{\sqrt{1 - v^2}} \qquad (c = 1). \tag{3.28}$$

Now compose a boost of velocity v with a boost of velocity u:

$$B(u)B(v) = \gamma_u \gamma_v \begin{pmatrix} 1 & -u \\ -u & 1 \end{pmatrix} \begin{pmatrix} 1 & -v \\ -v & 1 \end{pmatrix} \tag{3.29}$$

$$= \gamma_u \gamma_v \begin{pmatrix} 1 + uv & -(u + v) \\ -(u + v) & 1 + uv \end{pmatrix}. \tag{3.30}$$

Susskind's lecture does not insist on proving immediately that this composite matrix has exactly the standard boost form. Instead, he uses the easier observation: if it is a boost, then the effective velocity is read off from the ratio of the off-diagonal entry to the diagonal entry. Thus

$$w = \frac{u + v}{1 + uv} \qquad (c = 1). \tag{3.31}$$

Restoring the speed of light gives the standard relativistic addition law,

$$w = \frac{u+v}{1+uv/c^2}. \tag{3.32}$$

Newton would have written $w = u + v$. Einstein inserts the denominator.

The lecture immediately checks the moral of the formula. If u and v are both close to the speed of light, the numerator tries to become $2c$, but the denominator simultaneously approaches 2. The result does not exceed c. Even at the extreme edge, the composition of subluminal velocities remains subluminal.

3.6 Proper Velocity as a Four-Vector and as a Tangent Vector

At this point the lecture changes gears again. We have learned how coordinates transform; now we ask what kinds of physical quantities transform naturally with them. Susskind says explicitly that before speaking about velocity we should speak about four-vectors in general.

A four-vector v^μ is any object that transforms under Lorentz transformations in the same way the coordinate vector X^μ transforms. It has four components, not merely three. Its covariant form is obtained, exactly as before, by changing the signs of the spatial components:

$$v_\mu = (-v^1, -v^2, -v^3, v^0). \tag{3.33}$$

Its invariant norm is

$$v_\mu v^\mu = (v^0)^2 - (v^1)^2 - (v^2)^2 - (v^3)^2, \tag{3.34}$$

and for two four-vectors w^μ and v^μ, the invariant contraction is

$$w_\mu v^\mu = w^0 v^0 - w^1 v^1 - w^2 v^2 - w^3 v^3. \tag{3.35}$$

These are the relativistic analogs of the norm and dot product in ordinary Euclidean geometry. Their key property is invariance: the components change from frame to frame, but the contraction does not.

Only now does the lecture ask what relativistic velocity ought to be.

3.6.1 Question & Answer

Question. Why is the ordinary velocity $(dx/dt, dy/dt, dz/dt)$ not the right relativistic vector?

Answer. It is a perfectly good classical quantity, but it does not extend naturally to a four-vector. If we try to append a time component in the obvious way, we get

$$\frac{dt}{dt} = 1, \tag{3.36}$$

which is trivial. More importantly, the resulting object does not transform simply under Lorentz transformations. The lecture's point is not that ordinary velocity is meaningless, but that it is not the object that behaves most cleanly in relativistic geometry.

The cure is to divide not by coordinate time, which depends on the frame, but by proper time, which does not.

Thus the relativistic velocity is defined by

$$u^\mu = \frac{dX^\mu}{d\tau}. \tag{3.37}$$

The lecture calls this the proper velocity. The word "proper" refers to the parameter in the denominator: the derivative is taken with respect to proper time along the worldline.

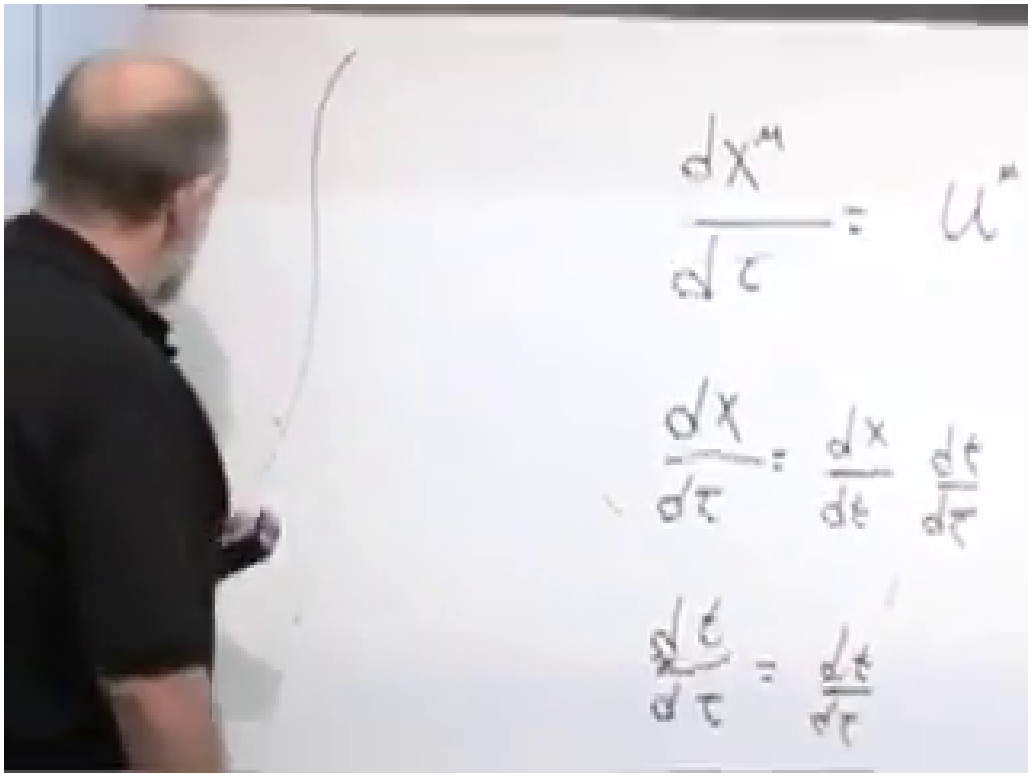


Figure 3.2: Four-velocity and the tangent-vector construction.

The board evidence is especially helpful here because it shows the defining equation placed beside a large curve on the board. The lecture is already telling us that the object is not merely algebraic. It is geometric.

The chain rule gives the first connection to the old three-velocity:

$$\frac{dx}{d\tau} = \frac{dx}{dt}\frac{dt}{d\tau}, \tag{3.38}$$

and similarly for the y- and z-components. The time component is simply

$$u^0 = \frac{dt}{d\tau}. \tag{3.39}$$

Thus everything depends on the single factor $dt/d\tau$.

To find it, start from the invariant interval:

$$d\tau^2 = dt^2 - dx^2 - dy^2 - dz^2. \tag{3.40}$$

Now divide by dt^2:

$$\left(\frac{d\tau}{dt}\right)^2 = 1 - \left(\frac{dx}{dt}\right)^2 - \left(\frac{dy}{dt}\right)^2 - \left(\frac{dz}{dt}\right)^2 \qquad (c = 1) \tag{3.41}$$

$$= 1 - \frac{v^2}{c^2}, \tag{3.42}$$

where in the second line we have restored c to make the units explicit. Hence

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{v^2}{c^2}}, \qquad \frac{dt}{d\tau} = \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}. \tag{3.43}$$

Substituting this back into the chain-rule relation gives the cleaned component form

$$u^i = \gamma v^i, \qquad u^0 = \gamma. \tag{3.44}$$

So the spatial pieces of the proper velocity are the ordinary velocity components multiplied by γ, and the time component is the same γ. At small velocities $\gamma \approx 1$, so $u^i \approx v^i$ and $u^0 \approx 1$. At relativistic velocities the difference becomes substantial, exactly as the lecture stresses.

The lecture then gives the geometric interpretation that makes this definition feel inevitable. In ordinary Euclidean geometry, if a curve has length element

$$ds^2 = dx^2 + dy^2 + dz^2, \tag{3.45}$$

then its unit tangent vector is obtained by dividing the differential displacement by the arc length:

$$\left(\frac{dx}{ds}, \frac{dy}{ds}, \frac{dz}{ds}\right). \tag{3.46}$$

This is a unit vector because

$$\left(\frac{dx}{ds}\right)^2 + \left(\frac{dy}{ds}\right)^2 + \left(\frac{dz}{ds}\right)^2 = 1. \tag{3.47}$$

The relativistic construction is the same idea with one crucial substitution: replace Euclidean arc length ds by proper time $d\tau$. Then

$$u^\mu = \frac{dX^\mu}{d\tau} \tag{3.48}$$

is the tangent vector to the worldline of the particle in spacetime.

This is where the lecture briefly pauses over terminology. In ordinary geometry, s is simply the distance measured along the curve. In relativity, the analogous parameter is $d\tau$, and because the interval carries the Lorentzian minus signs, one should not blur away the distinction between proper time and ordinary Euclidean distance merely by setting $c = 1$. The role is analogous; the geometry is not identical.

The next step then comes almost automatically. If u^μ is the tangent vector, its derivative with respect to proper time is the curvature data of the worldline:

$$a^\mu = \frac{du^\mu}{d\tau}. \tag{3.49}$$

This is the proper acceleration. The lecture does not yet build the full relativistic mechanics, but it clearly announces the direction: the analog of Newton's $F = ma$ must be phrased in terms of proper acceleration, not in terms of the second derivative with respect to coordinate time.

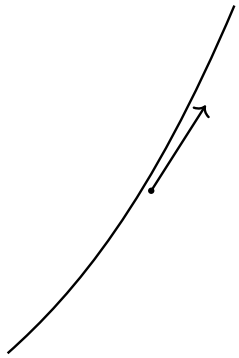


Figure 3.3: A schematic curve with its tangent vector, the Euclidean model for the relativistic construction.

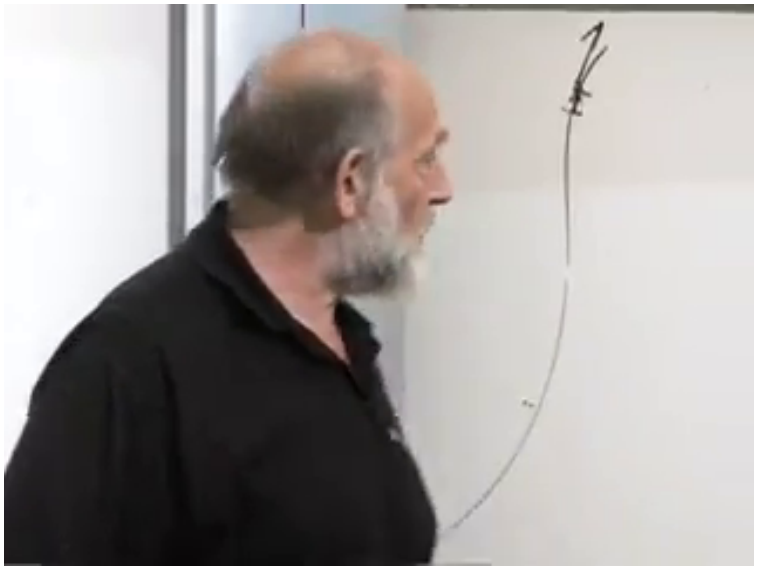


Figure 3.4: Curved trajectory for the transition from proper velocity to proper acceleration.

3.7 Momentum, Energy, and the Threshold of Mechanics

Having reached proper velocity, the lecture takes one further step into mechanics and then stops. The quantity chosen is momentum, because momentum is the basic conserved object in collision problems, and the lecture wants to hand that machinery to the next chapter rather than complete it here.

In Newtonian mechanics, one writes

$$\mathbf{p} = m\mathbf{v}. \tag{3.50}$$

Susskind pauses to stress a modern convention: mass is not treated as a velocity-dependent quantity. It is an intrinsic characteristic of the particle, what used to be called the rest mass. What changes relativistically is not the mass, but the velocity factor that must be used.

The correct relativistic definition is

$$p^\mu = mu^\mu. \tag{3.51}$$

The first three components are therefore

$$p^i = mu^i = \frac{mv^i}{\sqrt{1 - v^2/c^2}}. \tag{3.52}$$

For small velocities the denominator is essentially 1, so we recover the ordinary momentum $p^i \approx mv^i$. At larger velocities the momentum grows faster than the Newtonian formula would suggest.

The time component is the genuinely new piece:

$$p^0 = \frac{m}{\sqrt{1 - v^2/c^2}}. \tag{3.53}$$

Because the lecture is using t rather than ct as the time coordinate, the conventional energy is

$$E = c^2 p^0 = \frac{mc^2}{\sqrt{1 - v^2/c^2}}. \tag{3.54}$$

That is the careful version of the lecture's claim that the fourth component of momentum is energy: strictly speaking, one multiplies by c^2 to get the usual energy units.

Susskind then works this out in the low-velocity limit. Let

$$\epsilon = \frac{v^2}{c^2}, \tag{3.55}$$

so that

$$E = mc^2(1 - \epsilon)^{-1/2}. \tag{3.56}$$

Using the binomial approximation

$$(1 - \epsilon)^p \approx 1 - p\epsilon \tag{3.57}$$

for small ϵ, and taking $p = -1/2$, we obtain

$$(1 - \epsilon)^{-1/2} \approx 1 + \frac{1}{2}\epsilon. \tag{3.58}$$

Therefore

$$E \approx mc^2 \left(1 + \frac{1}{2}\frac{v^2}{c^2}\right) \tag{3.59}$$

$$\approx mc^2 + \frac{1}{2}mv^2. \tag{3.60}$$

The first term is the rest energy; the second is the ordinary Newtonian kinetic energy. In that way the time component of the four-momentum reproduces familiar mechanics in the nonrelativistic limit.

This is the lecture's final conceptual ascent. Energy and momentum are not two separate conserved quantities that merely coexist. They are components of a single four-vector. Conservation of four-momentum is conservation of momentum and energy together. More than that: under Lorentz transformations the components mix. What looks like pure energy in one frame becomes a combination of energy and momentum in another. That is why the relativistic language is the right one for collisions, decays, and mechanics in general.

The lecture stops there on purpose. Proper acceleration has been named, four-momentum has been defined, and the low-velocity limit has been tied back to Newtonian mechanics. The real use of these tools is postponed to the next stage.

3.8 Summary

The lecture begins with a five-dimensional picture of heavy-ion collisions and ends at the threshold of relativistic mechanics. That is not a loose collection of topics; it is a guided progression. First the black-hole analogy is stabilized and defended as either a genuine extra-dimensional description or a mathematically equivalent avatar of QCD. Then collider practice forces us to think about effective energy and radiation loss. Only after that physical motivation is in place does the lecture return to relativity.

From there the structure is exact and deliberate. The garage paradox isolates simultaneity as the hidden source of confusion. Covariant notation is then introduced as a disciplined way to write the invariant interval. Rotations provide the easy model for transformation laws, matrices make the structure manageable, and the composition of boosts yields the relativistic addition of velocities. Finally, ordinary velocity is rejected as the wrong relativistic object, proper velocity appears as the tangent vector to the worldline, and four-momentum packages momentum and energy into a single conserved whole. The lecture closes not with completion, but with a threshold: the machinery of relativistic mechanics is now in place, and the next step is to use it.

Chapter 4

Maxwell Equations, Covariance, and Electromagnetic Waves

We begin exactly where the previous lecture stopped. We already know how electric and magnetic fields act on charged particles; the Lorentz force law tells us that part of the story. The missing half is the reverse question: what determines the fields themselves, and how do charges and currents generate them? In this lecture Leonard Susskind answers that question in a deliberately relativistic way. Maxwell's equations are not introduced merely as a list to memorize, but as field equations to be motivated physically, written covariantly, checked component by component, and then used to derive the existence of light waves. Throughout we work in units with $c = 1$. These notes follow Leonard Susskind closely and are curated by LazyingArt LLC.

4.1 From the Lorentz Force to the Missing Field Equations

The lecture opens by looking backward before it moves forward. Electric and magnetic fields push charged particles around, and the force is proportional to the charge because an uncharged particle does not feel the electromagnetic field at all. In ordinary three-vector language the law is

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \tag{4.1}$$

This is the familiar half of electromagnetism: given the fields, we know how matter responds.

The board then pivots toward covariant notation. As written there, the second line is only schematic; it suppresses the free force index and the contraction with the four-velocity. The clean covariant statement is

$$f^\mu = q\, F^\mu{}_\nu\, u^\nu. \tag{4.2}$$

Here f^μ is the four-force, u^ν is the four-velocity, and $F^{\mu\nu}$ is the antisymmetric electromagnetic field tensor that we will reconstruct shortly. The point of the reminder is not just formal prettiness. Written this way, the Lorentz force law is manifestly the same in every inertial frame.

That is already the standard against which Maxwell's equations will be judged. The missing question is no longer how a charged particle moves in a prescribed field. The missing question is what determines the field itself: what decides whether there is an electric field here, a magnetic field

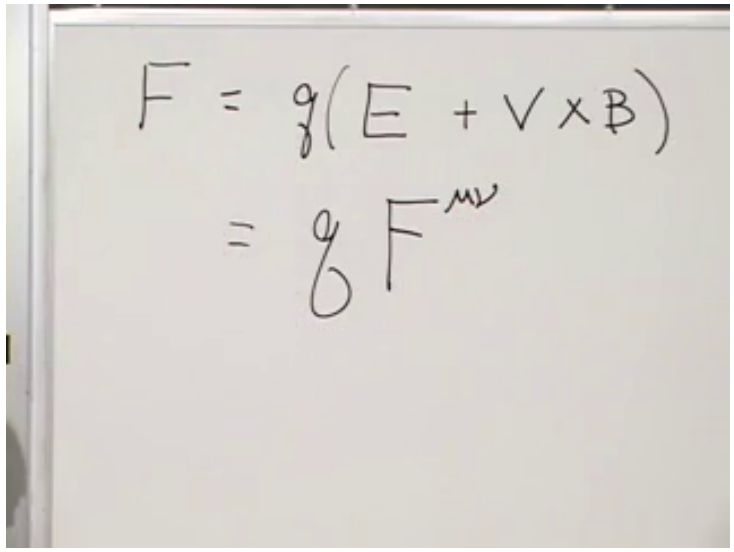


Figure 4.1: Lorentz force rewritten schematically in field-tensor notation.

there, or an electromagnetic wave moving through empty space? Those are the equations we now need, and those equations are Maxwell's equations.

As always in this course, the broader issue is relativity. If Maxwell's equations are the theory of light, and if the laws of physics are the same in all inertial frames, then light itself must come out with the same speed in all inertial frames. That is the puzzle Einstein struggled with, and the lecture keeps it in view from the start.

4.2 Why Charges Must Affect Fields

Before writing any field equations, Susskind slows down and asks why charged matter must affect the field at all. A point charge creates an electric field spreading outward from it. A current in a wire creates a magnetic field circulating around the wire. Those are familiar empirical facts, but the lecture wants the deeper logic that makes Maxwell's equations necessary.

The key point is reciprocity sharpened by conservation laws. An electromagnetic wave can hit a charged particle, change its motion, and give it energy. If that can happen, then the influence cannot run only one way. Charged matter must also be able to alter the electromagnetic field. Otherwise energy could appear in the particle without being taken from anywhere.

4.2.1 Question & Answer

Question. Why must charged matter act back on the electromagnetic field?

Answer. Because otherwise energy conservation would fail. If a field can do work on a charge, then a charge must be able to draw energy from the field and modify it. One may call this action and reaction, but the lecture's sharper point is conservation: the field cannot give energy to matter unless matter can affect the field in return.

The lecture then turns that abstract point into a causal picture. Suppose we move a point charge suddenly from one place to another. The field cannot rearrange itself instantaneously throughout all of space. A distant observer is not allowed to see the effect before a light signal could have arrived. Thus the field must rearrange by sending out a disturbance. Near the charge, the rearrangement happens first; farther away, the old field remains until the signal reaches it. If we shake the charge back and forth, the outward-moving disturbance becomes an electromagnetic wave. An antenna is precisely such a device.

The same story can be told for currents. Reverse a current in a wire, and the magnetic field around the wire must reverse as well. But that reversal cannot be known arbitrarily far away in arbitrarily short time. The change in the magnetic field must propagate outward, again as an electromagnetic disturbance. This is the basic physics Maxwell's equations must describe.

4.3 Wave Review and Vector-Calculus Review

Only now does the lecture pause for review. This is not filler. Before Maxwell's equations can be made to produce waves, we first recall what wave equations look like.

Start with a scalar field ϕ, a single number attached to each point of spacetime. The relativistic wave equation is

$$\partial_\mu \partial^\mu \phi = 0. \tag{4.3}$$

With the sign convention used throughout the lecture, this means

$$\partial_t^2 \phi - \partial_x^2 \phi - \partial_y^2 \phi - \partial_z^2 \phi = 0. \tag{4.4}$$

If the wave moves along the z-axis, then ϕ depends on z and t but not on x or y, and the equation reduces to

$$\partial_t^2 \phi - \partial_z^2 \phi = 0. \tag{4.5}$$

The lecture emphasizes the two basic traveling-wave solutions,

$$\phi = f(z - t), \qquad \phi = g(z + t), \tag{4.6}$$

representing right-moving and left-moving profiles. A particularly useful special case is the sinusoidal wave

$$\phi(z, t) = \cos\big(k(z - t)\big), \tag{4.7}$$

where k is the wave number. Large k means short wavelength; small k means long wavelength. These are exactly the shapes that will later reappear in the electric and magnetic fields themselves.

The lecture then reminds us of the vector operations Maxwell's equations use. For ordinary three-vectors,

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z, \tag{4.8}$$

$$(\mathbf{a} \times \mathbf{b})_x = a_y b_z - a_z b_y, \tag{4.9}$$

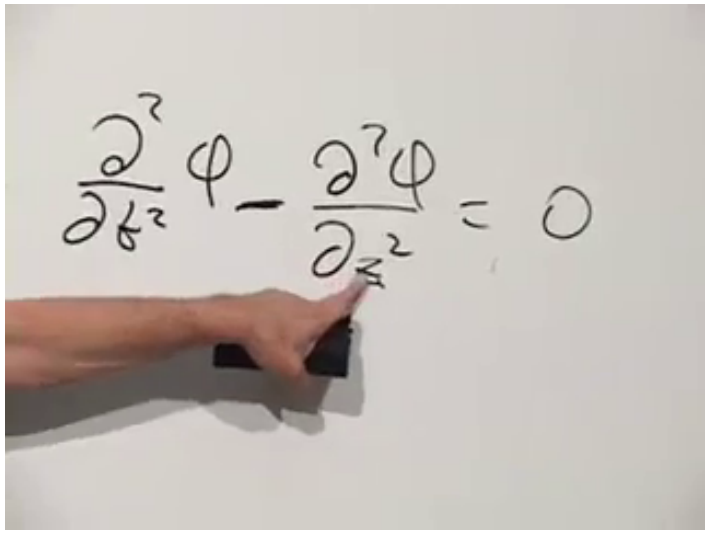


Figure 4.2: One-dimensional reduction of the wave equation.

with the remaining components obtained by cyclic permutation. For a vector field $\mathbf{A}$,

$$\nabla \cdot \mathbf{A} = \partial_x A_x + \partial_y A_y + \partial_z A_z, \tag{4.10}$$

$$(\nabla \times \mathbf{A})_x = \partial_y A_z - \partial_z A_y, \tag{4.11}$$

again with cyclic permutation for the other components.

Susskind keeps the geometry in sight. Divergence measures the extent to which a field seems to emanate from somewhere. Curl measures the extent to which it circulates around something. A field of arrows flowing outward from a point suggests divergence; a field winding around a wire suggests curl. These are the operational meanings needed when Maxwell's equations are written down.

4.4 Vacuum Maxwell Equations and the Relativity Puzzle

With that preparation in place, the lecture writes the source-free Maxwell equations, in the order actually used on the board:

$$\nabla \times \mathbf{E} = -\,\partial_t \mathbf{B}, \tag{4.12}$$

$$\nabla \cdot \mathbf{B} = 0, \tag{4.13}$$

$$\nabla \times \mathbf{B} = \partial_t \mathbf{E}, \tag{4.14}$$

$$\nabla \cdot \mathbf{E} = 0. \tag{4.15}$$

These are the vacuum Maxwell equations. They describe electric and magnetic fields in the absence of charge density and current density.

Two features are immediately visible. First, the curl equations are almost symmetric under interchange of $\mathbf{E}$ and $\mathbf{B}$, but not quite: there is an essential sign difference. Second, the divergence equations are asymmetric. The electric field will later acquire a source term ρ, but the magnetic field does not acquire an analogous source term in this lecture, so $\nabla \cdot \mathbf{B} = 0$.

Susskind briefly points out where sources will eventually go. One will later write $\nabla \cdot \mathbf{E} = \rho$ and $\nabla \times \mathbf{B} = \partial_t \mathbf{E} + \mathbf{J}$, but for the moment those terms are set aside. This postponement is deliberate. The vacuum equations reveal the covariant structure in its cleanest form.

4.4.1 Question & Answer

Question. How can Maxwell's equations be the same in every frame if they imply the same light speed in every frame?

Answer. That is precisely the relativistic puzzle. If Maxwell's equations are the theory of light, and if those equations are the same in every inertial frame, then every inertial frame must predict the same light speed from them. The resolution is not to patch the wave speed by hand from frame to frame. The resolution is to express the laws in terms of objects with definite Lorentz transformation rules, so that the equations themselves are manifestly frame-independent.

That is why the lecture now turns to the electromagnetic field tensor. The payoff is not elegance by itself. The payoff is that once the equations are written as tensor equations, their invariance is visible, and the constancy of the speed of light follows from the same structure.

4.5 The Electromagnetic Field Tensor

The next move is the central formal pivot of the lecture. The electric and magnetic fields are assembled into a single antisymmetric 4×4 tensor $F^{\mu\nu}$. The time coordinate is placed first, so the row and column order is t, x, y, z. The board image fixes the electric entries and the antisymmetric layout.

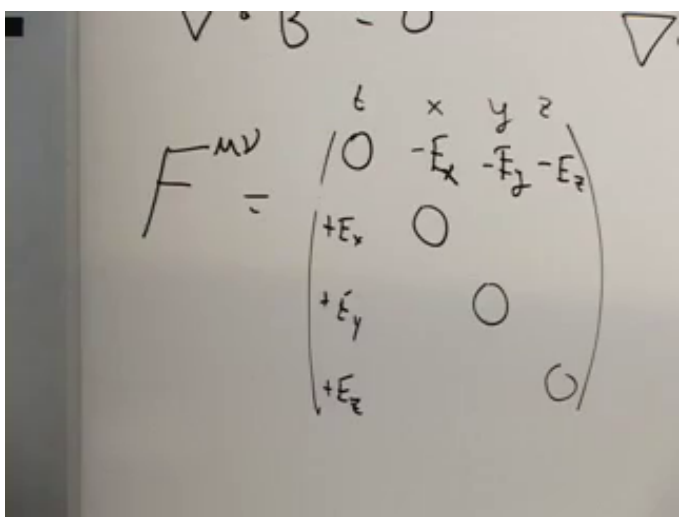


Figure 4.3: Board layout of the electromagnetic field tensor.

With the sign convention fixed by the visible board entries, we write

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & -B_z & B_y \\ E_y & B_z & 0 & -B_x \\ E_z & -B_y & B_x & 0 \end{pmatrix}. \tag{4.16}$$

The diagonal entries vanish because the tensor is antisymmetric, and the entries below the diagonal are the negatives of those above it. The time-space slots contain the electric field; the purely spatial slots contain the magnetic field.

The lecture then introduces a second antisymmetric tensor, the dual tensor $\tilde{F}^{\mu\nu}$. We do not need to overdevelop it. What matters is the replacement rule, stated cautiously because Susskind corrects himself while talking through the signs:

$$\mathbf{E} \mapsto -\mathbf{B}, \qquad \mathbf{B} \mapsto \mathbf{E}. \tag{4.17}$$

With that convention, $\tilde{F}^{\mu\nu}$ packages the other half of Maxwell's equations.

At this stage the lecture's rhythm is very clear: write the tensor, inspect its slots, and then ask what covariant first-derivative equations it could possibly satisfy. Since Maxwell's equations contain only first derivatives and should leave one free index, the natural guess is

$$\partial_\mu F^{\mu\nu} = 0, \tag{4.18}$$

$$\partial_\mu \tilde{F}^{\mu\nu} = 0. \tag{4.19}$$

There are four equations in the first line and four in the second, one for each choice of ν.

Now comes the trust-building step: we check components. For the time component,

$$\partial_\mu F^{\mu t} = \partial_t F^{tt} + \partial_x F^{xt} + \partial_y F^{yt} + \partial_z F^{zt} \tag{4.20}$$

$$= \partial_x E_x + \partial_y E_y + \partial_z E_z \tag{4.21}$$

$$= \nabla \cdot \mathbf{E}. \tag{4.22}$$

Thus $\partial_\mu F^{\mu t} = 0$ is exactly $\nabla \cdot \mathbf{E} = 0$.

To stay close to the lecture, it is also useful to check a spatial component. Taking $\nu = z$,

$$\partial_\mu F^{\mu z} = \partial_t F^{tz} + \partial_x F^{xz} + \partial_y F^{yz} + \partial_z F^{zz} \tag{4.23}$$

$$= -\partial_t E_z + \partial_x B_y - \partial_y B_x \tag{4.24}$$

$$= -\partial_t E_z + (\nabla \times \mathbf{B})_z. \tag{4.25}$$

So $\partial_\mu F^{\mu z} = 0$ is precisely the z-component of

$$\nabla \times \mathbf{B} = \partial_t \mathbf{E}. \tag{4.26}$$

The other spatial components work the same way. Likewise, $\partial_\mu \tilde{F}^{\mu\nu} = 0$ packages $\nabla \cdot \mathbf{B} = 0$ together with $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$.

This is the formal heart of the lecture. Once Maxwell's equations are written as tensor equations, we can see immediately that they are the same in every inertial frame.

4.6 From Maxwell to Electromagnetic Waves

The lecture then pivots from invariance to content: now that Maxwell's equations are written covariantly, what do they actually say? The answer is that they allow wave-like solutions.

We begin with

$$\nabla \times \mathbf{B} = \partial_t \mathbf{E} \tag{4.27}$$

and differentiate once more with respect to time:

$$\partial_t^2 \mathbf{E} = \nabla \times \partial_t \mathbf{B}. \tag{4.28}$$

Now we use the other curl equation,

$$\partial_t \mathbf{B} = -\nabla \times \mathbf{E}, \tag{4.29}$$

to eliminate $\mathbf{B}$. This gives

$$\partial_t^2 \mathbf{E} = -\nabla \times (\nabla \times \mathbf{E}). \tag{4.30}$$

So far the equations still involve the whole vector $\mathbf{E}$. The next step, exactly as in the lecture, is to isolate one component, say E_x.

The board algebra is long, but its strategy is simple. The x-component of the double curl can be reorganized into a clean form:

$$\partial_t^2 E_x = -\big(\nabla \times (\nabla \times \mathbf{E})\big)_x \tag{4.31}$$

$$= -\partial_x(\nabla \cdot \mathbf{E}) + \partial_x^2 E_x + \partial_y^2 E_x + \partial_z^2 E_x. \tag{4.32}$$

In vacuum, $\nabla \cdot \mathbf{E} = 0$, so the first term drops out and we obtain

$$\partial_t^2 E_x - \partial_x^2 E_x - \partial_y^2 E_x - \partial_z^2 E_x = 0. \tag{4.33}$$

Thus the x-component of the electric field satisfies the same wave equation that we earlier wrote for the scalar field ϕ. This is the exact mathematical payoff Susskind wants. The electric field itself can propagate as a wave.

A simple plane-wave example is now immediate:

$$E_x(z,t) = \cos(z-t), \qquad E_y = 0. \tag{4.34}$$

This is a right-moving wave along the z-axis in the lecture's units. It is also polarized: the electric field has been chosen to point along the x-direction. With the sign convention fixed above, a consistent accompanying magnetic field is

$$B_y(z,t) = \cos(z-t), \qquad B_x = B_z = 0. \tag{4.35}$$

This explicit B_y is not copied from the board; it is the clean consequence of the chosen sign convention together with $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$.

4.6.1 Question & Answer

Question. Why must the electromagnetic wave be transverse?

Answer. Because the divergence and curl equations leave no room for a propagating longitudinal component. For a wave depending only on z and t,

$$\nabla \cdot \mathbf{E} = \partial_x E_x + \partial_y E_y + \partial_z E_z = \partial_z E_z. \tag{4.36}$$

In vacuum this must vanish, so

$$\partial_z E_z = 0. \tag{4.37}$$

Therefore E_z can at most be a constant. A constant field is not part of the wave profile, so the wave part of the z-component must be absent.

Now look at the curl. If $E_x(z,t)$ is nonzero while $E_y = 0$ and the wave part of E_z vanishes, then the only nonzero component of $\nabla \times \mathbf{E}$ lies along y. But $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$, so the magnetic field itself must point along y. Hence

$$\mathbf{E} \perp \hat{z}, \qquad \mathbf{B} \perp \hat{z}, \qquad \mathbf{E} \perp \mathbf{B}. \tag{4.38}$$

That is the transversality of electromagnetic radiation. The electric field, the magnetic field, and the direction of propagation are mutually perpendicular.

Remark 4.1. The next frame is kept because it preserves the board geometry of the wave, even though its subtitle metadata is mismatched. Its evidentiary value here is visual: it belongs to the wave-and-polarization discussion, not to the anti-symmetric-matrix discussion suggested by the subtitle timing.

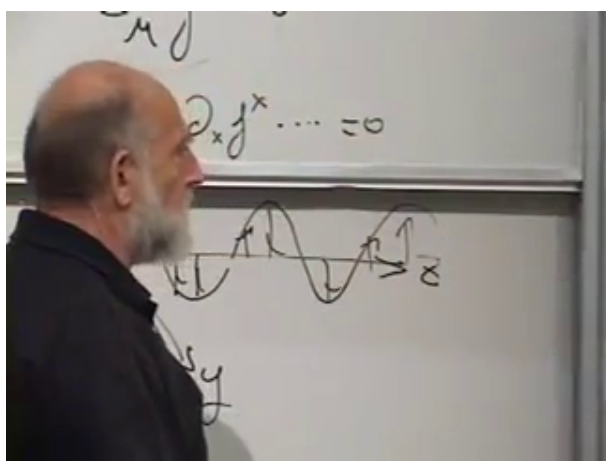


Figure 4.4: Board sketch of a transverse wave propagating along z.

This is the lecture's physical climax. Maxwell's equations imply electromagnetic waves, those waves move with speed 1 in the chosen units, and their electric and magnetic fields are transverse and polarized. Because the equations themselves are the same in every frame, these structural properties belong to every inertial observer.

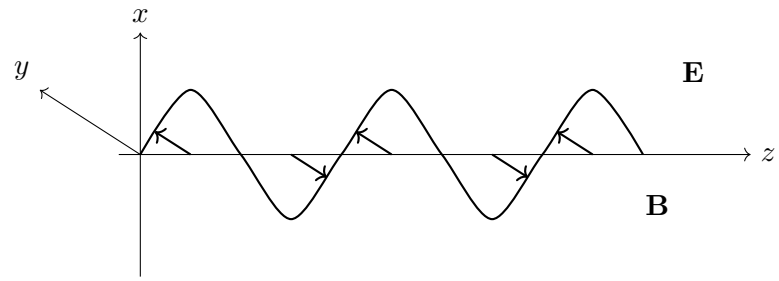


Figure 4.5: A clean reconstruction of the transverse electromagnetic-wave geometry: propagation along z, electric field along x, magnetic field along y.

4.7 Restoring Sources: Charge, Current, Continuity, and Covariance

Only after the vacuum story is clear does the lecture restore what had been postponed: charge density and current density. The Maxwell equations become

$$\nabla \cdot \mathbf{E} = \rho, \tag{4.39}$$

$$\nabla \cdot \mathbf{B} = 0, \tag{4.40}$$

$$\nabla \times \mathbf{E} = -\,\partial_t \mathbf{B}, \tag{4.41}$$

$$\nabla \times \mathbf{B} = \partial_t \mathbf{E} + \mathbf{J}. \tag{4.42}$$

Here ρ is the charge density and $\mathbf{J}$ is the current density.

The charge density is defined by

$$\rho = \frac{dQ}{dV}, \tag{4.43}$$

so that the charge in a region V is

$$Q_V = \int_V \rho \, dV. \tag{4.44}$$

The lecture emphasizes the continuum approximation: in reality space may contain many individual charges, but if we coarse-grain over sufficiently small regions we may treat charge as a continuous density field.

The density ρ can depend on time as well as position, because charge can move. But it cannot change arbitrarily. The total amount of charge in the universe is conserved, and the lecture immediately strengthens that statement. Charge is not allowed simply to disappear here and reappear there with nothing in between. If charge leaves one region and appears in another, there must be a flow of charge connecting the two.

4.7.1 Question & Answer

Question. Why is charge conservation stronger than the statement that total charge is constant?

Answer. Because global constancy by itself would still allow absurd behavior: charge could vanish on Earth and reappear on Mars without anything happening in between. The lecture forbids that. Charge may move from one place to another only by flowing through the intervening space. That stronger, local form of conservation is what leads to current density and to the continuity equation.

To describe such flow, take an infinitesimal oriented surface element $d\boldsymbol{\sigma}$. The charge per unit time passing through it is

$$\frac{dQ}{dt} = \mathbf{J} \cdot d\boldsymbol{\sigma}. \tag{4.45}$$

This is why current is a vector. Its magnitude is charge per unit area per unit time, and its direction is the direction in which charge flows.

Now fix a spatial region V with outward-pointing boundary element $d\boldsymbol{\sigma}$. If net current flows outward through the boundary, the charge inside must decrease. Thus

$$\frac{dQ_V}{dt} = -\oint_{\partial V} \mathbf{J} \cdot d\boldsymbol{\sigma}. \tag{4.46}$$

Since $Q_V = \int_V \rho \, dV$, we also have

$$\frac{dQ_V}{dt} = \int_V \partial_t \rho \, dV. \tag{4.47}$$

Combining the two,

$$\int_V \partial_t \rho \, dV = -\oint_{\partial V} \mathbf{J} \cdot d\boldsymbol{\sigma}. \tag{4.48}$$

At this point the lecture invokes Gauss's theorem, also called the divergence theorem:

$$\oint_{\partial V} \mathbf{A} \cdot d\boldsymbol{\sigma} = \int_V (\nabla \cdot \mathbf{A}) \, dV. \tag{4.49}$$

Applying it to $\mathbf{J}$ gives

$$\int_V \partial_t \rho \, dV = -\int_V (\nabla \cdot \mathbf{J}) \, dV. \tag{4.50}$$

If this holds for every region V, then the integrands themselves must agree, and we obtain

$$\partial_t \rho + \nabla \cdot \mathbf{J} = 0. \tag{4.51}$$

This is the continuity equation, the local form of charge conservation.

Now the relativistic packaging becomes natural. The four quantities ρ, J_x, J_y, J_z form the four-current

$$J^\mu = (\rho, J_x, J_y, J_z), \tag{4.52}$$

and the continuity equation becomes

$$\partial_\mu J^\mu = 0. \tag{4.53}$$

The sourced Maxwell equations themselves then collapse into a single covariant equation:

$$\partial_\mu F^{\mu\nu} = J^\nu. \tag{4.54}$$

For $\nu = t$, the right-hand side is $J^t = \rho$, so the equation becomes

$$\nabla \cdot \mathbf{E} = \rho. \tag{4.55}$$

For spatial ν, it becomes

$$\nabla \times \mathbf{B} = \partial_t \mathbf{E} + \mathbf{J}. \tag{4.56}$$

The other half of Maxwell's equations remains

$$\partial_\mu \tilde{F}^{\mu\nu} = 0. \tag{4.57}$$

So the return of sources does not spoil covariance. On the contrary, it completes it. Charges and currents enter as a four-vector, and the field equations remain tensor equations. The way fields act on matter is Lorentz invariant, and the way matter generates fields is Lorentz invariant as well.

4.8 Summary

The lecture has a very definite arc. We begin with the Lorentz force law and identify the missing half of the subject: not how fields act on matter, but how matter determines fields. Maxwell's equations answer that question. In vacuum they can be written as the covariant equations $\partial_\mu F^{\mu\nu} = 0$ and $\partial_\mu \tilde{F}^{\mu\nu} = 0$, and from them we derive wave equations for the field components themselves. A plane-wave solution then shows that electromagnetic radiation is transverse and polarized, with $\mathbf{E}$, $\mathbf{B}$, and the direction of propagation mutually perpendicular.

Finally, when charges and currents are restored, the continuity equation and the four-current J^μ lead to the sourced covariant equation $\partial_\mu F^{\mu\nu} = J^\nu$. That is the closing point Susskind wants to leave with us: electromagnetism can be written entirely in Lorentz-covariant form, and because those same laws imply waves moving with speed c, light travels with the same speed in every inertial frame.